SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based Questions

COURSE NAME: COMPUTER VISION **COURSE CODE: ITA05**

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images.* (BTL 3)

QUESTIONS: 5 **Max Marks50** **Duration :60 Minutes**

Annexure A

Scenario-Based Questions

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Scenario-Based Questions

1. **Dental X-Ray Enhancement and Cavity Region Extraction**

A dental clinic uses a digital radiography system to help dentists locate early-stage cavities (caries) in teeth. The X-ray images are dark and low in contrast, with quantum noise and overlapping tooth structures, so small cavity regions blend into the surrounding enamel and dentine.

Question: Apply suitable grey-level transformations (such as negative, log or power-law transformation) and histogram equalization to improve the visibility of the cavity regions. Select an appropriate spatial smoothing filter to suppress noise without blurring the cavity boundaries. Subsequently, apply thresholding or region growing to extract the suspected cavity region, and use edge detection to outline it. Explain the sequence of processing steps and justify the choice of each technique.

2. **Night-Time Vehicle Number Plate Region Extraction**

A traffic police department installs CCTV cameras at a toll gate to capture vehicles at night. The images suffer from very low brightness, headlight glare, sensor noise, and motion blur. The system must locate and extract the number-plate region so that the characters can be read in a later stage.

Question: Apply suitable grey-level transformations and histogram processing to enhance the dark regions while controlling glare. Use appropriate smoothing and sharpening filters (for example, median filtering followed by Laplacian or unsharp masking) to reduce noise and restore character edges. Apply edge detection, line detection and edge linking to locate the rectangular plate boundary, and use thresholding to separate the characters from the plate background. Present the complete processing pipeline and explain how each step contributes to extracting the number-plate region.

3. **Textile Fabric Defect Detection Using Frequency-Domain Processing**

A textile mill inspects woven fabric rolls using a line-scan camera. The fabric has a regular, repeating weave pattern, and defects such as holes, broken threads, oil stains and knots disturb this pattern. The captured images also contain periodic noise from the machine lighting and uneven illumination across the width of the roll.

Question: Apply suitable frequency-domain filtering techniques (low-pass, high-pass, or notch/band-reject filtering) to suppress the periodic noise and the regular weave texture, so that the defects stand out. Use histogram processing to correct the uneven illumination. Subsequently, apply thresholding and boundary detection to segment and outline the defective regions. Explain the order of operations and justify why frequency-domain processing is preferred over spatial-domain processing in this scenario.

4. **Blood Cell Segmentation in Microscopic Images**

A pathology laboratory uses an automated microscope to capture images of stained blood smears for counting red and white blood cells. The images show uneven staining, poor contrast between cells and background, touching or overlapping cells, and noise from dust particles on the slide.

Question: Apply suitable contrast enhancement and histogram processing techniques to improve the separation between cells and background. Use an appropriate spatial filter to remove small noise particles while preserving cell boundaries. Apply global or adaptive thresholding to separate cells from the background, and use region-based segmentation (region growing or region splitting and merging) to separate touching cells. Finally, apply edge detection and boundary detection to outline individual cells. Explain the processing sequence and the role of each technique in obtaining accurate cell regions.

5. **Text Extraction from Degraded Historical Documents**

A state archive is digitising old handwritten manuscripts. The scanned pages contain yellowed paper, ink bleed-through from the reverse side, stains, faded strokes and uneven lighting across the page. The archive requires clean extraction of text regions before handwriting recognition.

Question: Apply suitable grey-level transformations and histogram processing to improve the contrast between faded ink and the background. Select appropriate smoothing and sharpening filters to remove stains and noise while strengthening the pen strokes. Compare global thresholding (such as Otsu's method) with local (adaptive) thresholding for this scenario and apply the more suitable one to separate text from the background. Use edge detection and boundary detection to identify text lines and word regions. Explain the sequence of operations and justify each decision.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand

Question 1: Dental X-Ray Enhancement and Cavity Region Extraction

1. Introduction

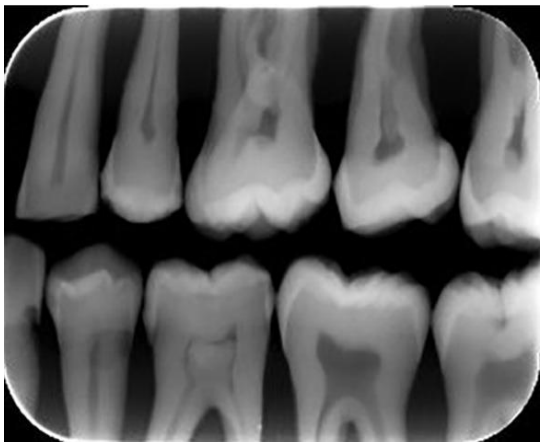
Dental X-rays help dentists identify cavities (dental caries) and tooth decay. However, digital X-ray images may have low brightness, poor contrast, quantum noise and overlapping tooth structures. These problems make small cavities difficult to distinguish from healthy enamel and dentine.

Computer vision techniques such as grey-level transformations, histogram equalization, spatial filtering, thresholding and edge detection can improve image quality and extract suspected cavity regions.

Objective: To enhance the dental X-ray, reduce noise while preserving cavity boundaries, segment suspected cavity regions and outline them using edge detection.

2. Grey-Level Transformations

Grey-level transformations modify pixel intensity values to improve the visibility of hidden details.



Original X-ray



Enhanced X-ray

A. Log transformation

$$s = c \log(1 + r)$$

Here, (r) is the input intensity, (s) is the output intensity and (c) is a scaling constant. Log transformation expands lower intensity values and compresses higher values, helping reveal subtle details in darker regions.

B. Power-law (gamma) transformation

$$s = cr^{\gamma}$$

- $\gamma < 1$:**brightens the image.**
- $\gamma > 1$:**darkens the image.**

A suitable gamma value below 1 may improve the visibility of dark details.

C. Negative transformation

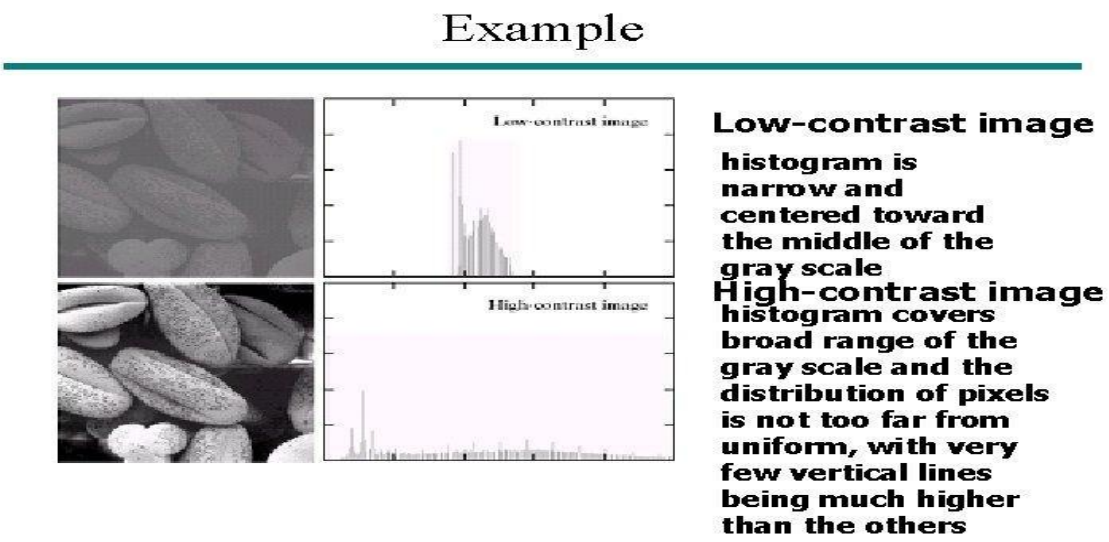
$$s = L - 1 - r$$

Here, (L) is the number of possible intensity levels. The transformation reverses the image intensities, which may make some structures easier to observe.

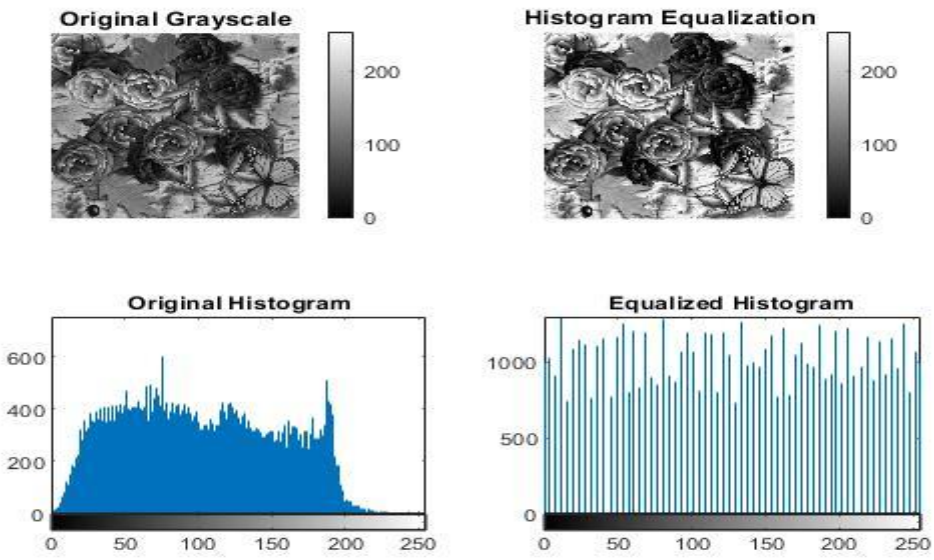
Decision: Select log or gamma transformation according to the original image brightness and contrast.

3. Histogram Equalization

Histogram equalization redistributes pixel intensities to improve image contrast.



Original histogram



After enhancement

Selected technique:

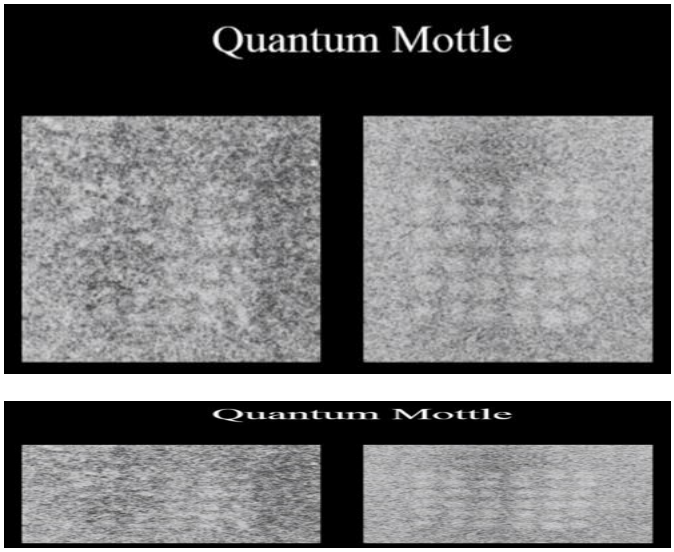
- CLAHE (Contrast Limited Adaptive Histogram Equalization).
- CLAHE enhances contrast in small image regions while limiting excessive amplification of noise.

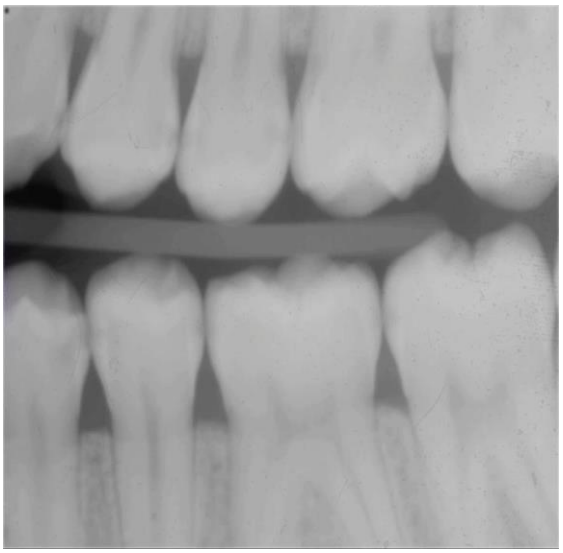
Advantages:

- Improves local contrast between tooth structures.
- Helps reveal subtle intensity variations.
- Makes possible cavity regions easier to examine.
- Limits excessive contrast amplification compared with unrestricted local enhancement.

4. Noise Removal

Quantum noise can hide fine details and cause incorrect segmentation. A spatial smoothing filter is therefore applied before extracting cavity regions.





Noisy image **Filtered image**

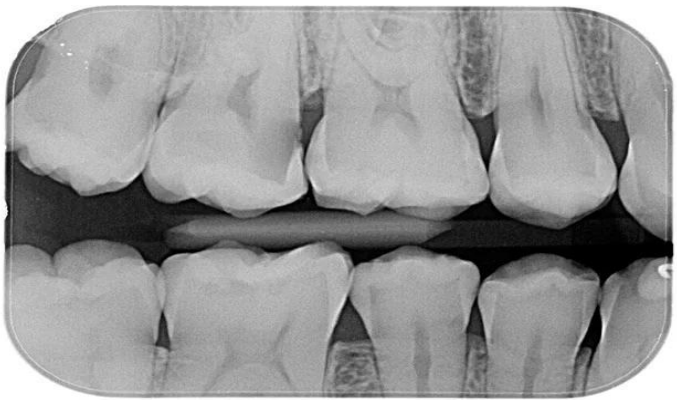
Selected technique: Median filtering.

A median filter replaces each pixel with the median value of neighbouring pixels, for example, within a 3×3

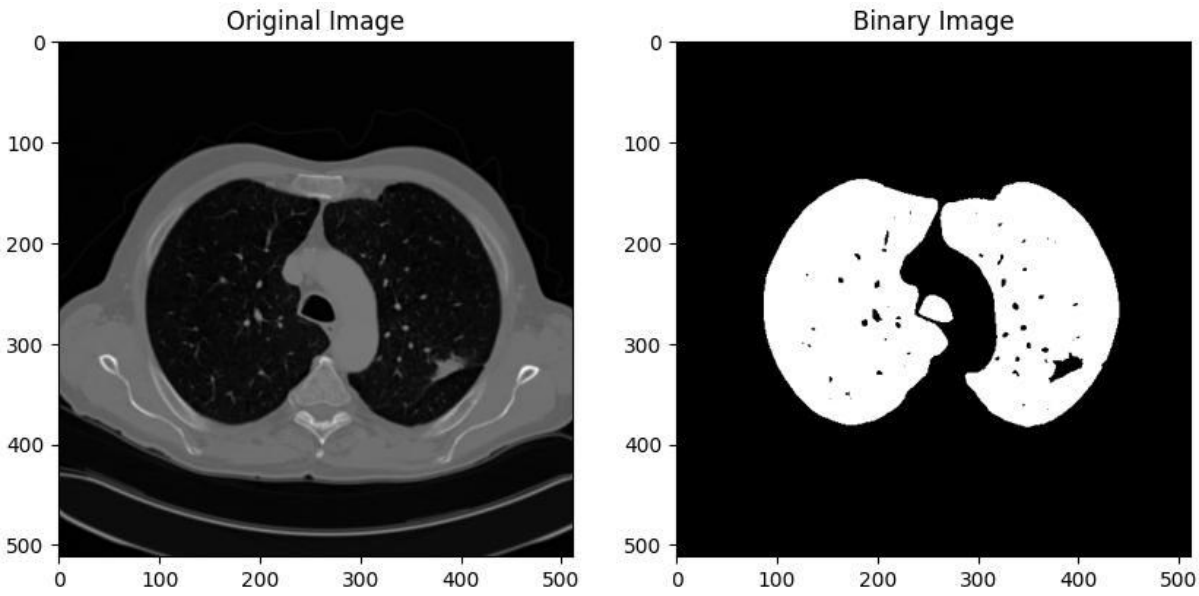
It reduces isolated noise while preserving important edges better than a simple averaging filter. A small kernel is preferred because excessive smoothing can remove small cavity details. A bilateral filter is another option when edge preservation is especially important**.**

5. Thresholding and Segmentation

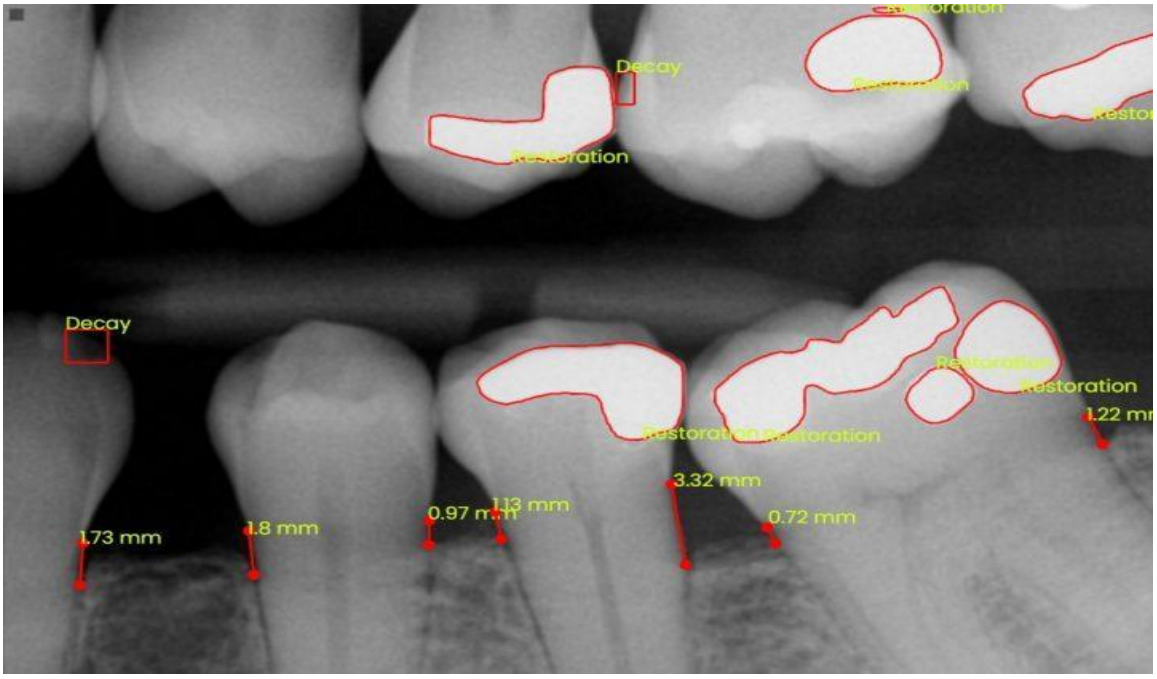
Segmentation separates suspected cavity regions from surrounding tooth structures.



Enhanced image



Thresholded image



Candidate region

A. Thresholding

Thresholding classifies pixels according to their intensity.

$$g(x,y) = \begin{cases} 255, & f(x,y) > T(x,y) \\ 0, & \text{otherwise} \end{cases}$$

Here, f(x,y)is the image intensity, T(x,y)is the threshold, and g(x,y) is the binary output. The inequality can be reversed to select darker regions.

Adaptive thresholding is useful when image contrast varies across different parts of the X-ray because it calculates a local threshold for each neighbourhood.

B. Region growing

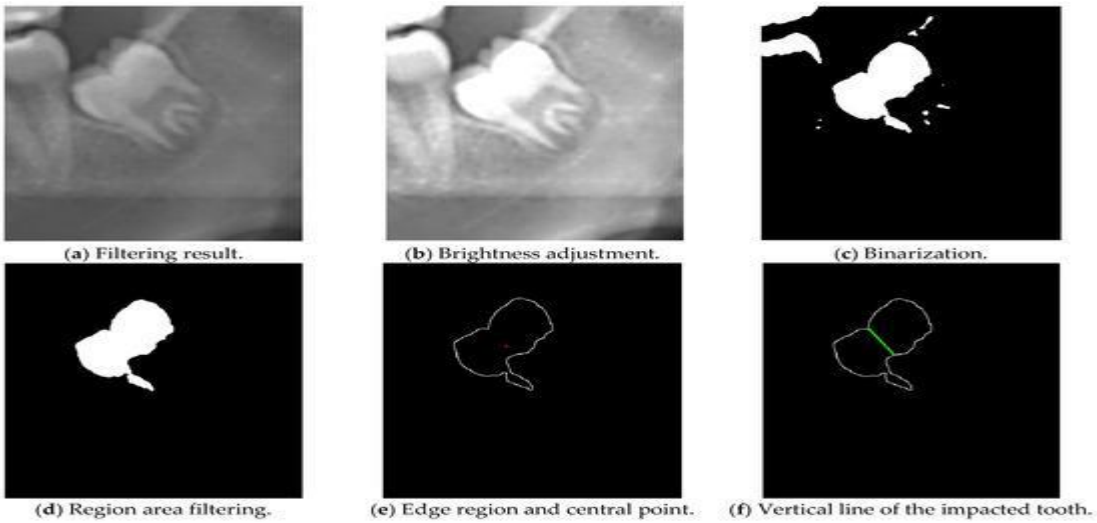
Region growing extracts connected regions based on similarity between neighbouring pixels.

Procedure:

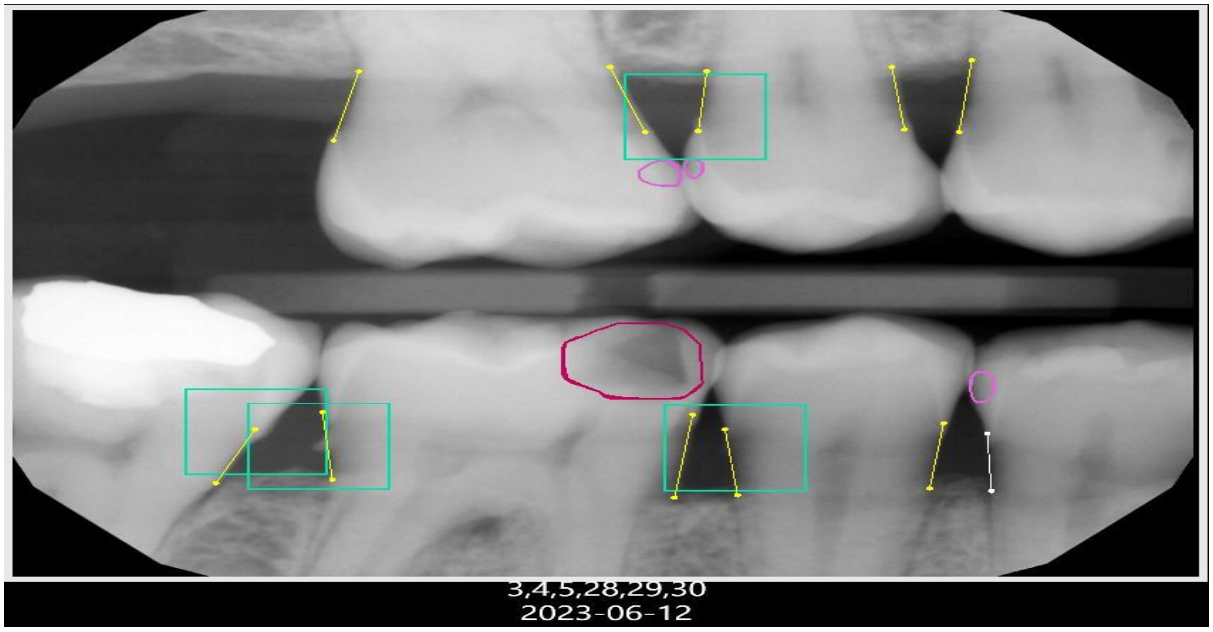
1. Select a seed pixel in a suspected cavity region.
2. Examine neighbouring pixels.
3. Add pixels that satisfy the selected intensity-similarity condition.
4. Continue until no suitable neighbouring pixels remain.
5. Remove irrelevant isolated regions using suitable morphological operations.

Decision: Use adaptive thresholding to generate candidate regions, followed by region growing to refine a selected region when intensity similarity is reliable.

6.Edge Detection and Boundary Extraction



Canny edge map



Outlined candidate region

The Canny edge detector identifies significant changes in image intensity and helps outline the segmented region.

Steps involved:

1. Smooth the image to reduce noise.
2. Calculate intensity gradients.
3. Apply non-maximum suppression to obtain thin edges.
4. Apply double-threshold hysteresis to retain meaningful edge connections.
5. Extract contours around the segmented candidate regions.

The final boundaries can be overlaid on the dental X-ray to support visual examination.

7.Overall Processing Flowchart:

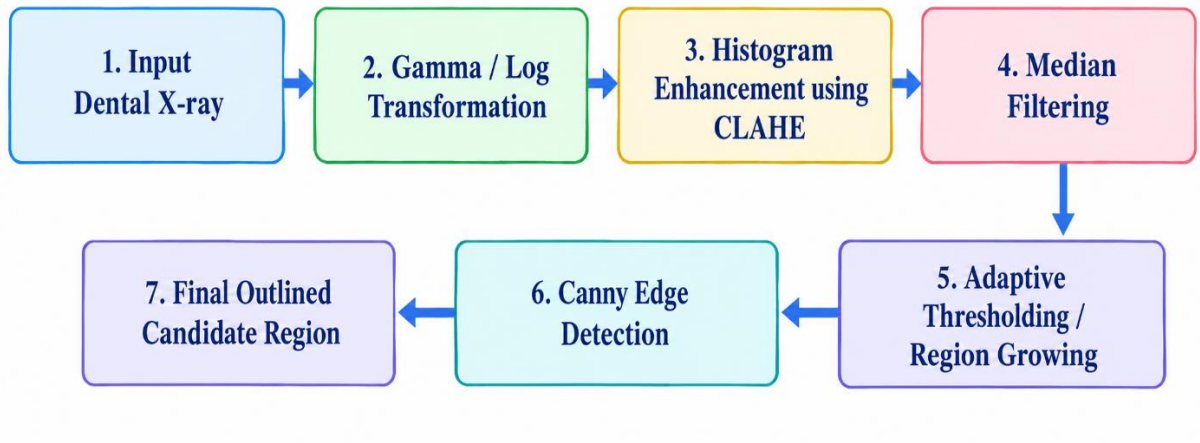


Figure 1: Dental X-ray Image Processing Flowchart.

8. Expected Results

Technique	Expected outcome
Grey-level transformation	Improved visibility of dark details
CLAHE	Better local contrast
Median filtering	Reduced noise
Thresholding	Binary candidate regions
Region growing	Refined connected regions
Canny detection	Detected edges
Contour extraction	Outlined candidate boundaries

9. Conclusion

The proposed processing pipeline combines grey-level transformations, CLAHE, median filtering, thresholding, region growing and Canny edge detection to improve the visibility of suspected dental cavities. Each technique addresses a specific image-quality problem and contributes to extracting candidate regions with clearer boundaries.

The final output is an enhanced dental X-ray with suspected cavity regions outlined for further clinical examination. The extracted regions must be validated by a dentist; image processing alone cannot confirm a diagnosis.

Question 2: Night-Time Vehicle Number Plate Region Extraction

1. Introduction

Automatic number plate recognition is an important computer vision application used by traffic police, toll gates, parking systems, and vehicle monitoring systems. At night, CCTV cameras often capture images with very low brightness, headlight glare, sensor noise, and motion blur. These problems make the number plate and its characters difficult to identify.

To solve this problem, a sequence of image-processing techniques is applied. Grey-level transformations improve brightness, histogram processing enhances contrast, smoothing filters reduce noise, sharpening techniques restore character edges, and edge and line detection methods locate the rectangular number plate. Finally, thresholding separates the characters from the plate background for further recognition.

Objective: To enhance a night-time vehicle image, locate the rectangular number plate, extract the plate region, and separate its characters while controlling noise, glare, and blur.

2. Identification of Key Issues

1. **Very low brightness:** The number plate appears dark and its characters are difficult to see.
2. **Headlight glare:** Bright headlights may cause overexposed regions and hide nearby details.
3. **Sensor noise:** Random pixel variations affect image quality.
4. **Motion blur**: Vehicle movement produces blurred character edges.
5. **Weak plate boundaries**: Low contrast makes the rectangular plate difficult to locate.
6. **Character-background similarity:** Characters may blend with the plate background, making thresholding difficult.

Therefore, the system requires controlled enhancement, noise reduction, edge restoration, rectangular boundary detection, and character segmentation.

3. Grey-Level Transformations

Grey-level transformations modify pixel intensities to improve the visibility of dark number plates and characters.



Original night image



Enhanced image

A.Logtransformation

$$s = c\log(1 + r)$$

Here, r is the input intensity, s is the output intensity, and c is a scaling constant.

Log transformation expands low-intensity values and compresses high-intensity values. It helps reveal details in dark regions without increasing bright values as aggressively as some other transformations.

B. Power-law (gamma) transformation

$$s = cr^{\gamma}$$

- $\gamma < 1$: $brightens\ dark\ regions.$
- $\gamma > 1$: $darkens\ the\ image.$

For a dark night-time image, gamma correction with $\gamma < 1$can improve plate visibility. However, it can also amplify noise and increase the apparent brightness of glare.

C. Negative transformation

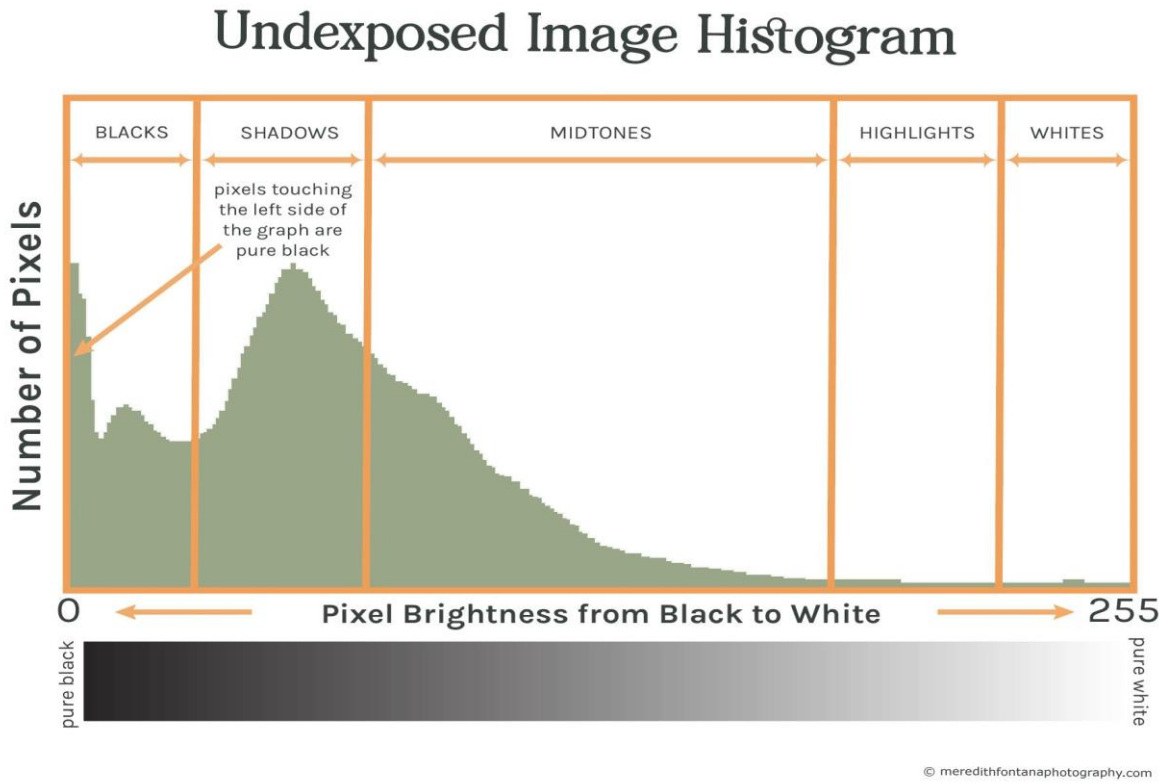
$$s = (L - 1) - r$$

This transformation reverses pixel intensities. It may help examine certain plate details, but it is not usually the first choice for number plate extraction because it also reverses the appearance of the entire scene.

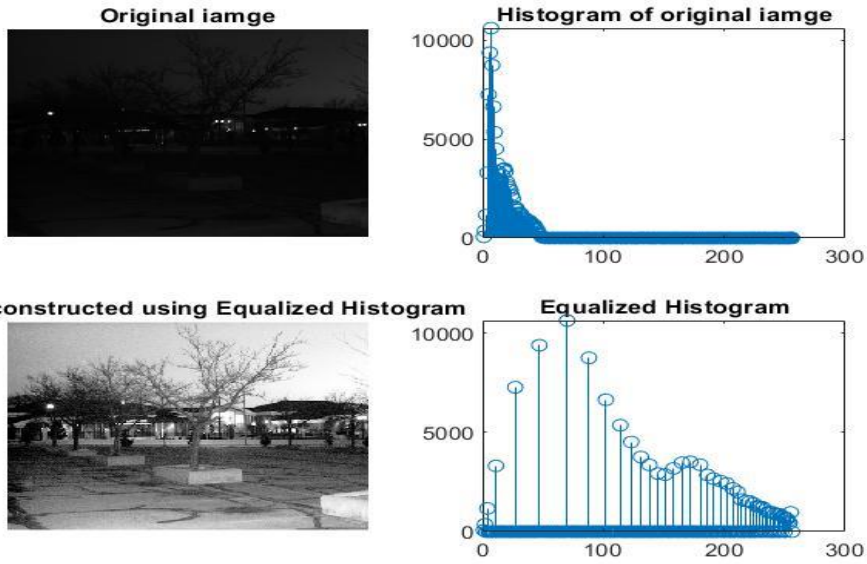
Decision: Apply gamma correction or log transformation first, and use glare-aware local enhancement to avoid excessively brightening headlights**.**

4. Histogram Processing and Glare Control

Histogram processing improves the distribution of image intensities and helps distinguish the plate from the surrounding vehicle.



Before enhancement



After enhancement

Selected technique: CLAHE (Contrast Limited Adaptive Histogram Equalization).

CLAHE enhances local contrast in different regions of an image while limiting excessive contrast amplification.

Advantages:

- Reveals details in dark number plate regions.
- Improves contrast between characters and plate background.
- Enhances local boundaries around the plate.
- Limits excessive amplification of noise.

To control glare, bright saturated areas should be identified separately, and enhancement should be limited in those regions. CLAHE cannot recover details that have been completely lost due to overexposure.

Decision: Use gamma correction followed by CLAHE with suitable contrast limits. If glare is severe, apply a glare mask or highlight suppression before further processing.

5. Noise Removal and Sharpening

Noise reduction should be performed before sharpening because sharpening can amplify unwanted pixel variations.



Noisy plate image



Filtered and sharpened plate

A. Median filtering

A median filter replaces each pixel with the median intensity of its neighbouring pixels. A small 3×3kernel is a reasonable starting point.

It reduces isolated noise while preserving edges better than simple averaging in many situations.

B. Laplacian sharpening

The Laplacian detects rapid intensity changes in an image.

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

A sharpened image can be calculated as:

$$g = f - \lambda \nabla^2 f$$

Here, f is the original image,g is the sharpened image, and λ controls sharpening strength under this convention.

Laplacian sharpening can enhance the edges of characters and plate boundaries. However, excessive sharpening may amplify noise and create false edges.

C. Unsharp masking

Unsharp masking creates a blurred version of the image, subtracts it from the original, and adds the resulting detail back.

$$g = f + k(f - f_{\text{blur}})$$

Here, k controls the sharpening strength.

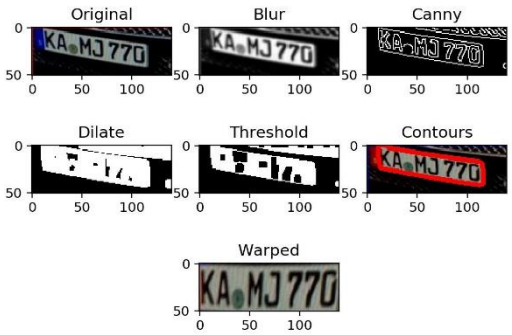
Decision: Apply median filtering first, followed by mild unsharp masking or Laplacian sharpening. Sharpening can improve apparent edge clarity, but it cannot fully restore details lost through severe motion blur.

6. Edge Detection and Plate Boundary Localization

After enhancement and filtering, edge detection identifies the intensity transitions that may correspond to the number plate's rectangular border.



Enhanced plate area



Edge map

Detected plate



Detected plate

A. Canny edge detection

Canny edge detection identifies significant intensity changes through smoothing, gradient calculation, non-maximum suppression, and hysteresis thresholding.

It is suitable for detecting:

- Horizontal plate borders.
- Vertical plate borders.
- Character strokes.
- Other strong structural edges in the image.

B. Line detection

The Hough line transform can identify straight lines from the detected edges. The number plate is expected to have approximately horizontal top and bottom borders and vertical side borders, allowing for perspective distortion.

The algorithm searches for groups of aligned edge pixels and identifies candidate plate boundaries.

C. Edge linking

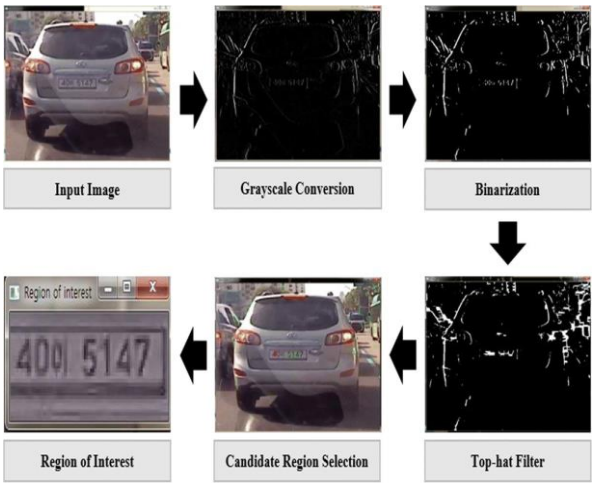
Edge linking connects nearby edge segments that are likely to belong to the same boundary. It helps bridge small gaps caused by noise, weak contrast, or partial obstruction.

Decision: Use Canny edge detection followed by Hough line detection and edge linking. Select a rectangular candidate using its geometry, aspect ratio, edge density, and location within the vehicle image.

7. Thresholding and Character Extraction

Once the number plate is located, crop the rectangular plate region and convert it into a suitable grayscale image. Thresholding then separates the characters from the plate background.





Cropped number plate **Binary characters**

Thresholding can be represented as:

$$g(x,y) = \begin{cases} 255, & f(x,y) > T \\ 0, & \text{otherwise} \end{cases}$$

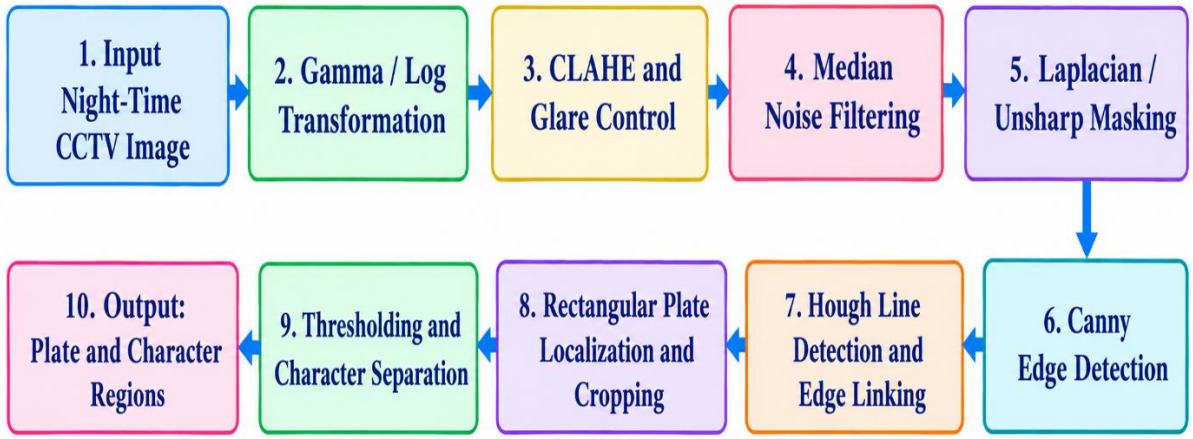
Here, f(x,y) is the pixel intensity, Tis the threshold, and g(x,y) is the binary image.

- **Global thresholding:** Uses one threshold for the entire cropped plate.
- **Adaptive thresholding:** Uses local thresholds and is more suitable when illumination varies across the plate.

The binary image may be inverted depending on whether the characters are darker or brighter than the background. Morphological opening or closing can remove small artifacts or reconnect broken character strokes, provided these operations do not merge adjacent characters.

Decision: Use adaptive thresholding when glare and uneven lighting cause intensity variations; use global thresholding when the cropped plate has consistent illumination and clear contrast.

8. Complete Processing Pipeline



9. Expected Results

Technique	Expected outcome
Gamma/log transformation	Better visibility of dark regions
CLAHE and glare control	Improved local contrast with controlled highlights
Median filtering	Reduced sensor noise
Sharpening	More distinct character edges
Canny detection	Detection of strong boundaries
Hough transform	Candidate horizontal and vertical plate lines
Edge linking	More continuous rectangular boundaries
Thresholding	Separation of characters and background

10. Conclusion

The proposed pipeline combines grey-level transformations, histogram processing, median filtering, sharpening, edge detection, line detection, edge linking, and thresholding to extract number plates from night-time CCTV images.

Gamma correction and CLAHE improve the visibility of dark regions, while glare control prevents excessively bright areas from dominating the enhancement. Median filtering reduces noise, and mild sharpening improves character-edge clarity. Canny detection, Hough line detection, and edge linking help locate the rectangular plate boundary. Finally, adaptive or global thresholding separates the characters from the plate background.

The resulting cropped plate and binary character image can be passed to an optical character recognition (OCR) system for reading the registration number. The extraction quality depends on image resolution, glare severity, and motion blur.

Question 3: Textile Fabric Defect Detection Using Frequency-Domain Processing

1. Introduction

Textile industries use computer vision to inspect fabric rolls and identify manufacturing defects such as holes, broken threads, oil stains, and knots. A line-scan camera captures fabric images while the material moves through the inspection system.

However, regular repeating weave patterns, periodic noise from machine lighting, and uneven illumination make defect detection difficult. Frequency-domain processing helps identify and suppress repeating patterns, allowing unusual fabric regions to stand out.

Objective: To suppress periodic noise and regular weave texture, correct uneven illumination, segment defective regions**, and outline their boundaries using suitable image-processing techniques.**

2. Identification of Key Issues

1. **Regular weave texture:** Repeating horizontal and vertical threads may hide small defects.
2. **Periodic noise:** Machine lighting can introduce repeating intensity variations.
3. **Uneven illumination:** Some fabric regions appear brighter or darker than others.
4. **Fabric defects**: Holes, broken threads, oil stains, and knots disturb the normal texture.
5. **Low defect contrast:** Certain defects have intensities similar to the surrounding fabric.
6. **Boundary detection errors**: Remaining noise and texture can produce false edges.

Therefore, the system needs frequency analysis, selective filtering, illumination correction, thresholding, and boundary extraction.

3. Proposed Processing Pipeline

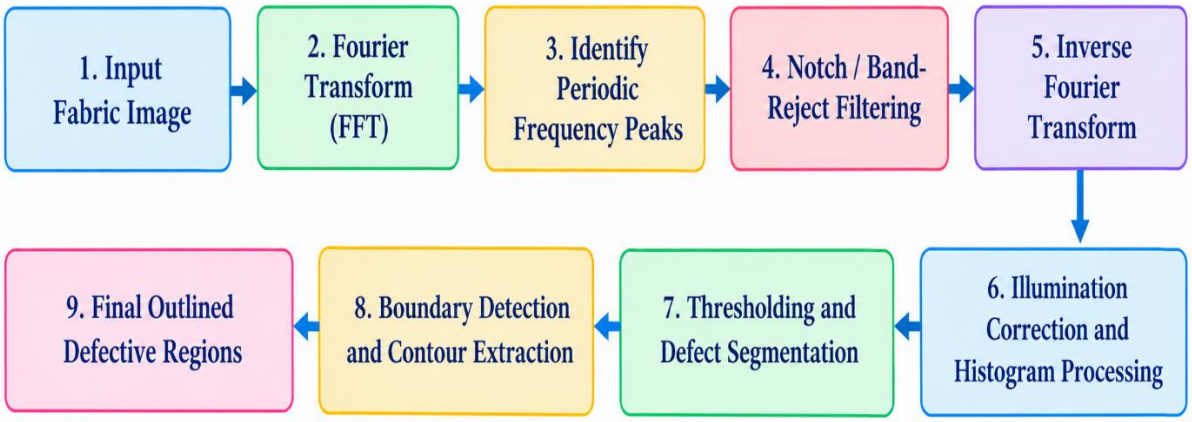


Figure 1: Fabric Defect Detection Flowchart Using FFT.

4. Frequency-Domain Processing

Frequency-domain processing represents an image as a combination of spatial frequencies. Low frequencies generally describe slowly varying intensity and broad structures, while high frequencies represent rapid changes such as thread edges and fine texture.

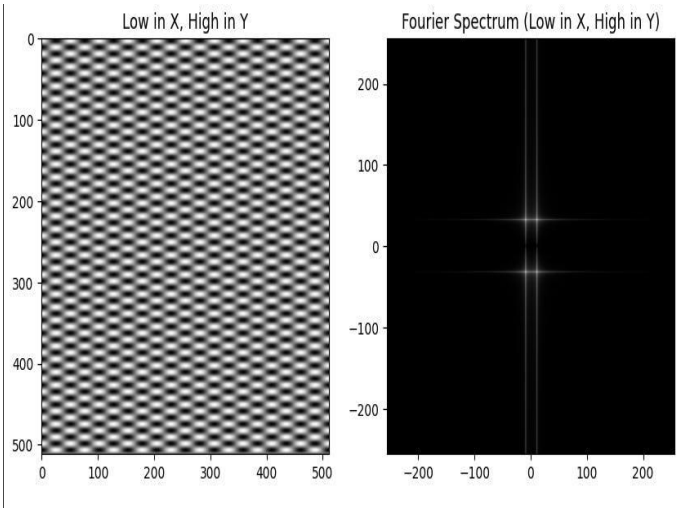
The two-dimensional Fourier transform is:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) e^{-j2\pi(ux/M+vy/N)}$$

- $f(x,y) is\ the\ original\ fabric\ image.$
- $F(u,v)\ is\ its\ frequency\ representation.$
- $u, v\ represent\ spatial - frequency\ coordinates.$
- M, N are the image dimensions.

A Fourier magnitude spectrum can reveal periodic components as bright peaks away from the central low-frequency region.





Fabric texture **Frequency spectrum**

5. Selecting the Appropriate Frequency Filter

A. Notch or band-reject filtering

Notch filters suppress specific frequency peaks associated with periodic lighting noise. Band-reject filters suppress a selected range of frequencies.

A frequency-domain filter is applied as

$$G(u, v) = H(u, v)F(u, v)$$

Here, H(u,v) is the filter transfer function, F(u,v) is the Fourier transform, and G(u,v) is the filtered spectrum.

Decision: Use notch filtering when the unwanted periodic noise appears as distinct peaks. Use band-reject filtering when the unwanted pattern occupies a broader frequency band.

B. Low-pass filtering

Low-pass filtering retains lower frequencies and suppresses high-frequency variations. It can reduce fine noise, but excessive filtering may blur holes, broken threads, and other small defects.

C**. High-pass filtering**

High-pass filtering emphasizes rapid intensity changes and can highlight broken threads or sharp defect boundaries. However, it may also amplify noise and the normal weave texture.

Decision: Notch or band-reject filtering is the primary choice for identifiable periodic noise. Low-pass and high-pass filters should be used selectively after considering the frequency content of the defects.

6. Inverse Fourier Transform and Histogram Processing

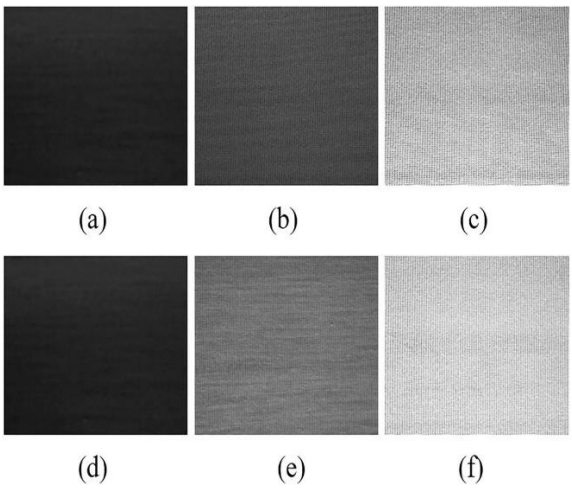
After filtering, the inverse Fourier transform converts the filtered spectrum back into an image:

$$g(x,y) = \mathcal{F}^{-1}\{G(u,v)\}$$

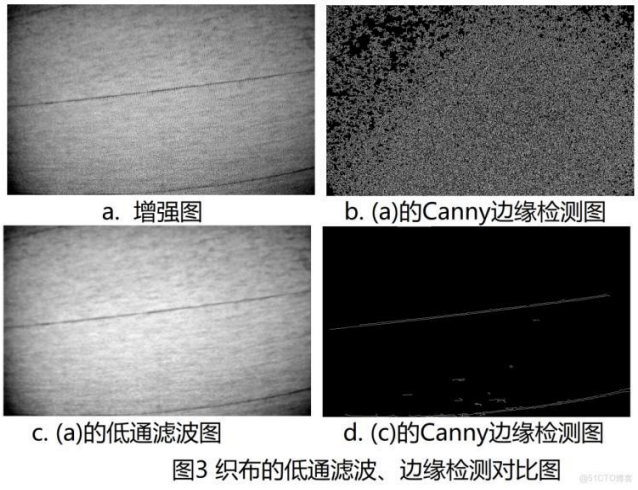
The resulting image should contain less periodic interference while preserving useful defect details.

The next step is illumination correction. A slowly varying background can be estimated and removed or divided out to reduce broad brightness variations across the fabric width.

Histogram processing can then improve contrast. CLAHE (Contrast Limited Adaptive Histogram Equalization) is useful for enhancing local contrast while limiting excessive amplification of noise.



Uneven illumination



Corrected illumination

Justification: Background correction reduces false detections caused by lighting variations, while CLAHE helps make subtle differences between normal fabric and defective regions more visible.

7. Thresholding for Defect Segmentation

Thresholding separates possible defective pixels from the surrounding fabric based on intensity.

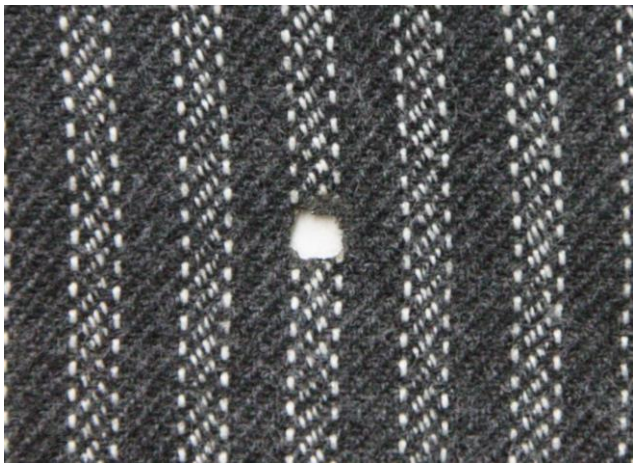
$$g(x,y) = \begin{cases} 255, & f(x,y) > T \\ 0, & \text{otherwise} \end{cases}$$

Here, f(x,y) is the corrected image intensity, Tis the threshold, and g(x,y) is the binary output. The inequality can be reversed when defects appear darker than the background.

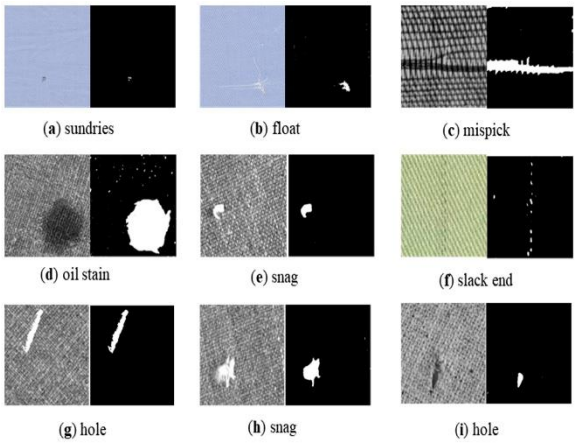
Global thresholding: Uses a single threshold for the entire image. It works best when illumination is reasonably uniform and defect contrast is consistent.

Adaptive thresholding: Calculates a local threshold for each neighbourhood. It is useful when illumination and defect contrast vary across the fabric.

Decision: Apply adaptive thresholding when some parts of the fabric remain brighter or darker than others. Use global thresholding if illumination correction produces a sufficiently uniform image.



Corrected fabric



Binary segmentation

8. Boundary Detection and Contour Extraction

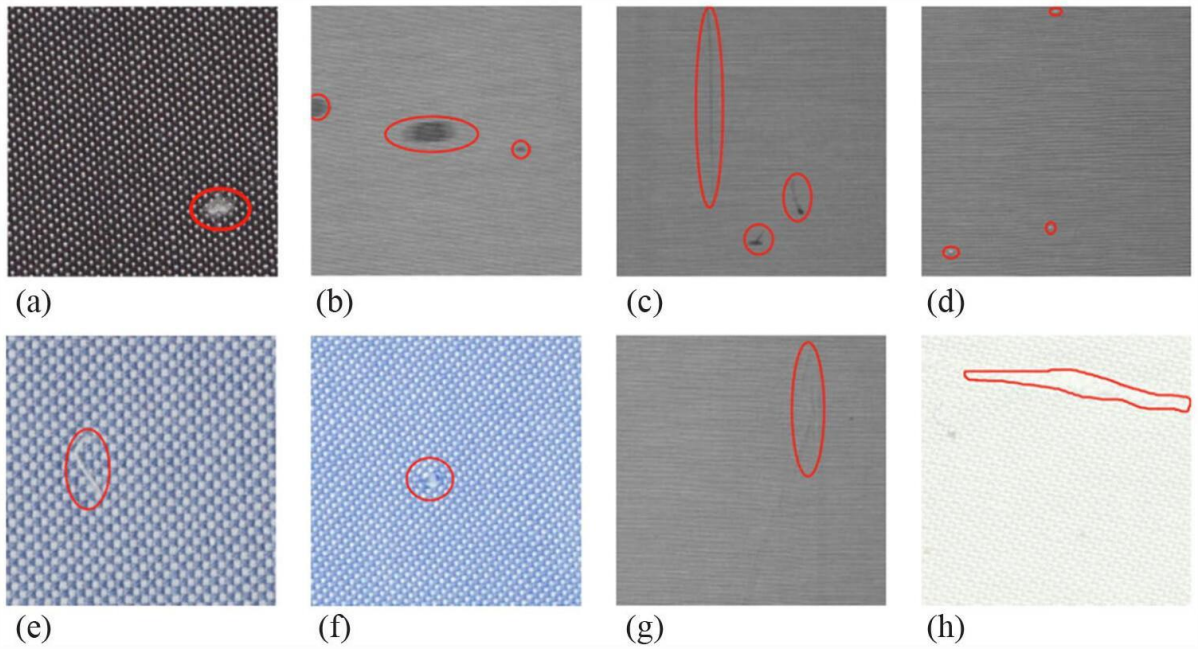
After segmentation, boundary detection identifies the outlines of the defective regions.

Procedure:

1. Generate a binary image using the selected threshold.
2. Apply suitable morphological opening to remove isolated noise.
3. Use morphological closing to fill small gaps when appropriate.
4. Detect boundaries using contour extraction or an edge detector such as Canny.
5. Measure region area, shape, and position to reject obvious false detections.
6. Overlay the detected contours on the original fabric image.



Defective region



Outlined defect

The final output shows the location and boundary of candidate fabric defects, enabling the inspection system to flag damaged portions of the fabric roll.

9. Why Frequency-Domain Processing Is Preferred

Frequency-domain processing	Spatial-domain processing
Analyzes repeating patterns through their frequency components.	Works directly on individual pixels and local neighbourhoods.
Can target distinct periodic noise peaks using notch filters.	Simple smoothing filters may not selectively remove a periodic component.
Can suppress selected weave frequencies when they are distinguishable from defect frequencies.	Local filters may blur defects together with the regular texture.
Useful when lighting interference is periodic and stable.	Useful for local operations such as median filtering, thresholding and morphology.

Important limitation: The regular weave and defects may share some frequencies. Removing too much of the spectrum can also remove genuine defect information. Therefore, the filter should be designed from the observed spectrum and validated on representative fabric images.

10. Expected Results

Processing stage	Expected result
Fourier transform	Reveals periodic frequency components
Notch / band-reject filter	Reduces selected periodic interference
Inverse Fourier transform	Produces the filtered fabric image
Illumination correction	Reduces broad brightness variations
CLAHE	Improves local contrast
Thresholding	Separates candidate defect pixels
Boundary detection	Outlines defective regions

11. Conclusion

The proposed method combines Fourier analysis, selective frequency-domain filtering, illumination correction, histogram enhancement, thresholding, and boundary detection to identify defects in woven fabric.

Frequency-domain processing is particularly suitable because periodic lighting noise and repeating weave patterns may produce identifiable frequency components. Notch or band-reject filters can suppress these components while retaining useful defect information. Illumination correction and CLAHE then improve the image for segmentation, and contour extraction outlines the candidate defective regions.

The final result is an inspected fabric image with suspected holes, broken threads, stains, or knots highlighted for quality-control review. Filter settings must be validated to ensure that genuine defects are not removed along with the repeating texture.

4)Blood Cell Segmentation in Microscopic Images

1. Introduction

Blood cell segmentation is an important application of digital image processing used in pathology laboratories to identify and separate red blood cells (RBCs) and white blood cells (WBCs) from microscopic blood smear images. It helps medical professionals count blood cells, measure their size, study their shape, and identify abnormal cell structures.

In an automated system, a microscope captures stained blood samples, and computer vision techniques process the images to identify individual cells. However, uneven staining, poor contrast, touching or overlapping cells, and noise caused by dust particles can affect the accuracy of segmentation. Therefore, image enhancement and filtering techniques are applied before separating the cells from the background.

Objective: To improve image contrast, remove unwanted noise, distinguish blood cells from the background, separate touching cells, and outline individual cell boundaries for accurate counting and analysis.

2. Identification of Key Issues

- **Uneven staining**: Different regions of the blood smear may have different brightness levels, making cell identification difficult.
- **Poor contrast:** The intensity of blood cells may be similar to that of the background, reducing visibility.
- **Touching cells**: Two or more cells may appear as one connected region and require further segmentation.
- **Overlapping cells**: Cell boundaries may be hidden where cells overlap.
- **Noise**: Dust particles and unwanted image variations may be incorrectly detected as cells.
- **Boundary errors**: Poor contrast and noise may produce incomplete or false cell outlines.

These problems can reduce the accuracy of cell counting and measurement. The system therefore requires contrast enhancement, histogram processing, spatial filtering, thresholding, region-based segmentation, and boundary detection.

3. Proposed Processing Pipeline

The following processing sequence improves the microscopic image step by step.

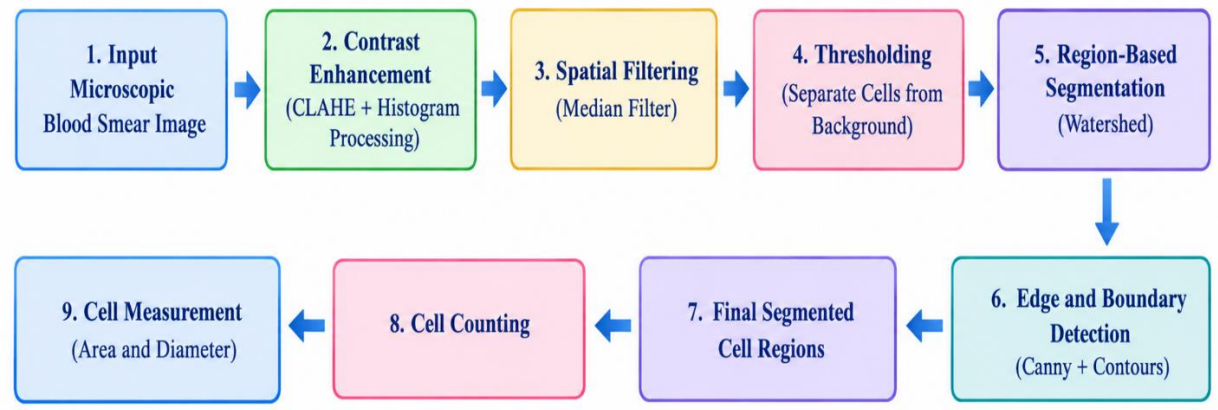


Figure 1: Microscopic Blood Cell Segmentation Flowchart.

Role: Each stage prepares the image for the next operation, helping the system produce more accurate cell regions and boundaries.

4. Contrast Enhancement and Histogram Processing

Contrast enhancement improves the difference between blood cells and the surrounding background. It is particularly useful when the microscopic image contains faint cells, uneven staining, or poor illumination.

Techniques:

- **Histogram equalization:** Redistributes pixel intensity values to improve the overall contrast of the image.

- **Contrast stretching**: Expands the intensity range so that faint cell structures become more visible.
- **CLAHE (Contrast Limited Adaptive Histogram Equalization):** Enhances local contrast in small image regions. It is useful for unevenly stained blood smears because it limits excessive amplification of noise.

A histogram represents the frequency distribution of pixel intensity values.

$$\boldsymbol{h(r_k) = n_k}$$

Where:

- r_k represents an intensity level.
- $n_k$represents the number of pixels having that intensity.
- $h(r_k)$ represents the frequency of that intensity level.

Role: Histogram processing improves the visibility of blood cells, helps distinguish cell regions from the background, and makes thresholding more effective.

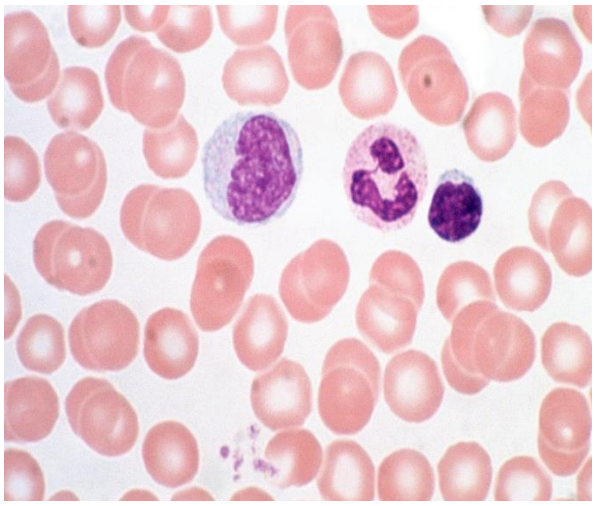
5. Spatial Filtering for Noise Removal

Spatial filtering processes pixels using their neighbouring pixel values to reduce unwanted noise while preserving important cell structures. It is applied after contrast enhancement to avoid detecting dust particles as actual blood cells.

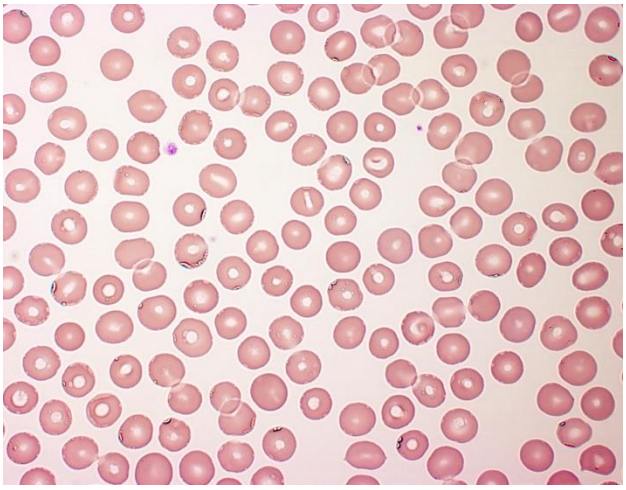
Filter	Function
Median filter	Removes isolated dust particles and salt-and-pepper noise while preserving edges.
Gaussian filter	Smooths fine image variations but may blur small cell boundaries.
Bilateral filter	Reduces noise while preserving strong intensity changes around cell boundaries.

Decision: A median filter is suitable for removing isolated noise particles. If additional smoothing is needed, a bilateral filter can be applied to preserve cell edges more effectively than ordinary Gaussian smoothing.

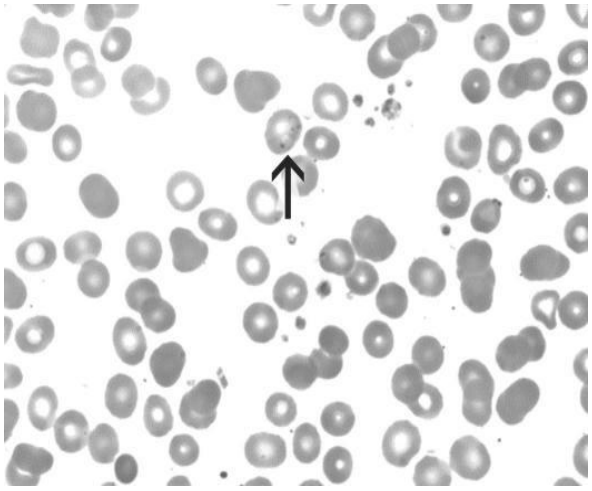
Role: Spatial filtering improves image quality, reduces false detections, and prepares the image for reliable thresholding and cell segmentation.



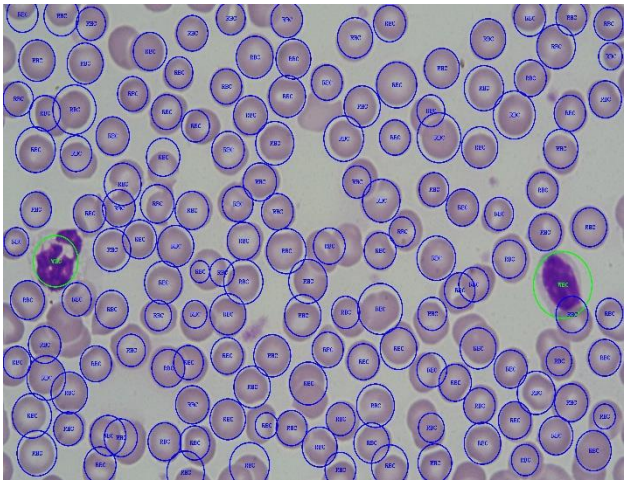
Original Image



Enhanced Image



Noisy Image



Filtered Image

6. Global and Adaptive Thresholding

Thresholding converts a grayscale microscopic image into a binary image to separate blood cells from the background. Each pixel is classified according to its intensity compared with a selected threshold value.

The thresholding equation is:

$$g(x,y) = \begin{cases} \mathbf{255}, & f(x,y) > T \\ \mathbf{0}, & \textbf{otherwise} \end{cases}$$

Where:

- $f(x,y) represents\ the\ input\ image\ intensity.$
- T represents the selected threshold value.
- $g(x,y)$ represents the binary output image.
- 255 represents white pixels, and 0 represents black pixels.

The inequality can be reversed when the cells are darker than the background.

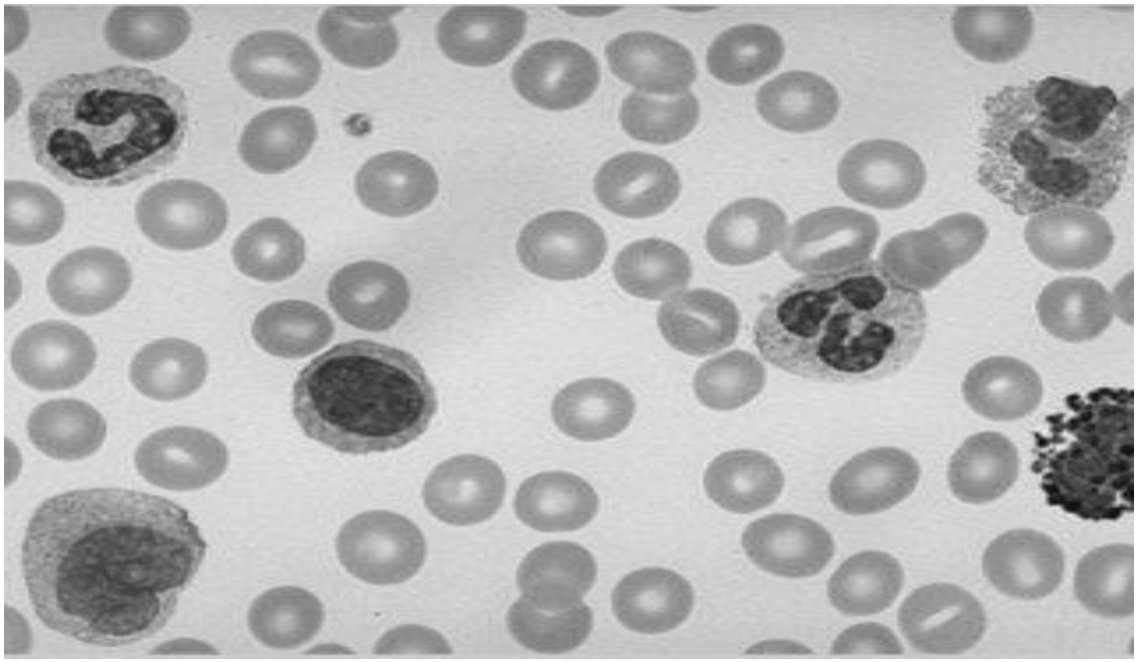
Techniques:

- Global thresholding (Otsu's method): Selects a single threshold for the entire image. It works well when the background illumination is reasonably uniform and cell contrast is consistent.
- Adaptive thresholding: Calculates a local threshold for each neighbourhood. It is useful when staining and illumination vary across different regions of the blood smear.

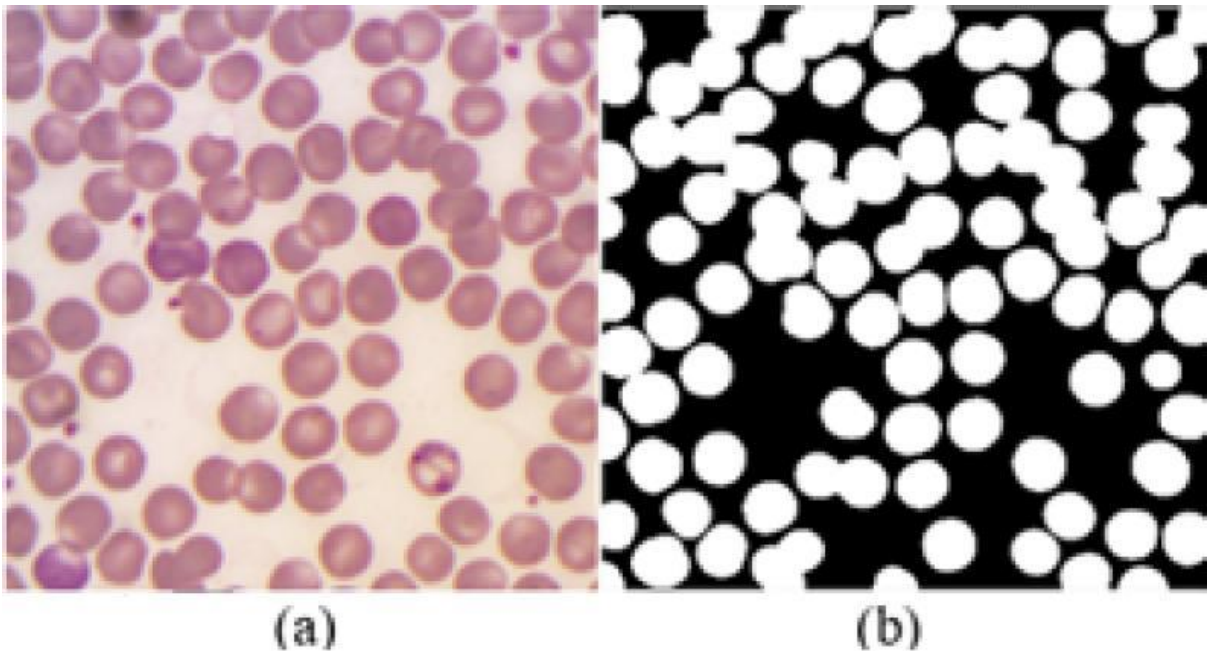
Decision: Adaptive thresholding is preferred when uneven staining remains after enhancement. Global thresholding is suitable when the image has uniform illumination and a clear intensity difference between cells and background.

Role: Thresholding creates a binary mask that identifies candidate cell regions for further segmentation.

Diagram 3 – Thresholding



Grayscale Image



Binary Threshold Image

7. Region-Based Segmentation

After thresholding, touching or overlapping blood cells may appear as one connected region. Region-based segmentation separates these connected regions into individual cells based on their intensity, colour, connectivity, or spatial properties.

Techniques:

- **Region growing:** Starts from a seed pixel and groups neighbouring pixels with similar intensity or colour.
- **Region splitting and merging:** Divides large regions into smaller sections and merges neighbouring regions with similar properties**.**
- **Watershed segmentation:** Uses a distance transform and markers to separate touching cells into individual regions.

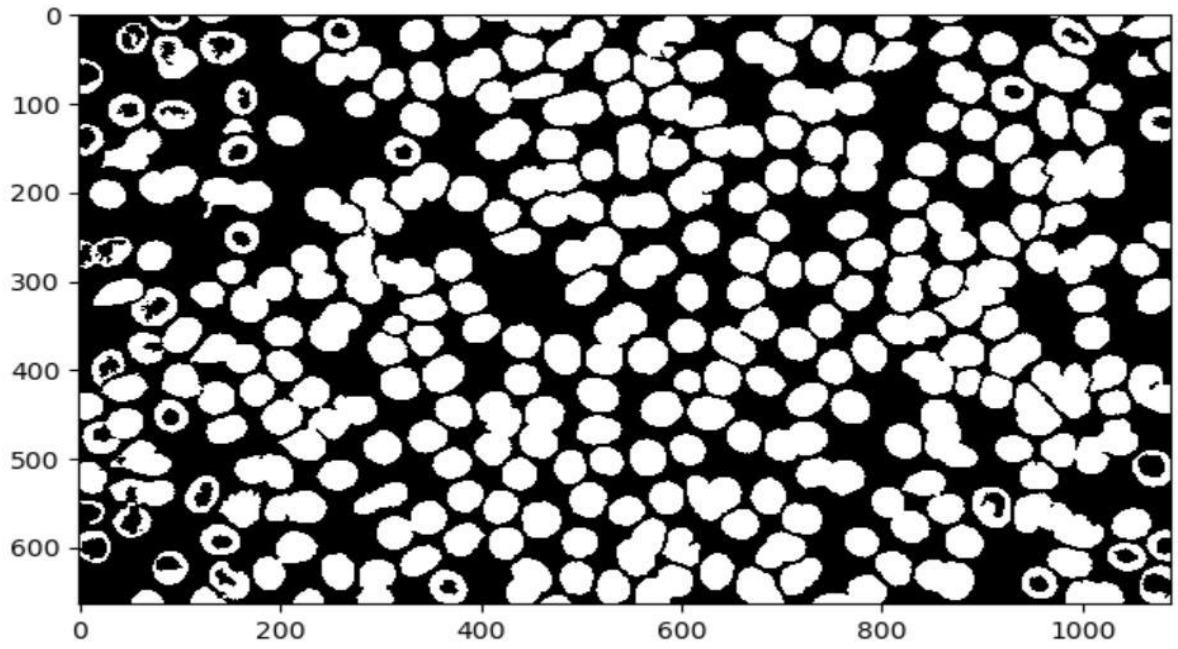
Procedure:

1. Identify foreground cell regions from the binary image.
2. Calculate the distance transform to locate the approximate centres of cells.
3. Mark individual cell centres as seed regions.
4. Apply marker-controlled watershed segmentation to separate touching cells.
5. Remove small unwanted regions that are likely to be noise.

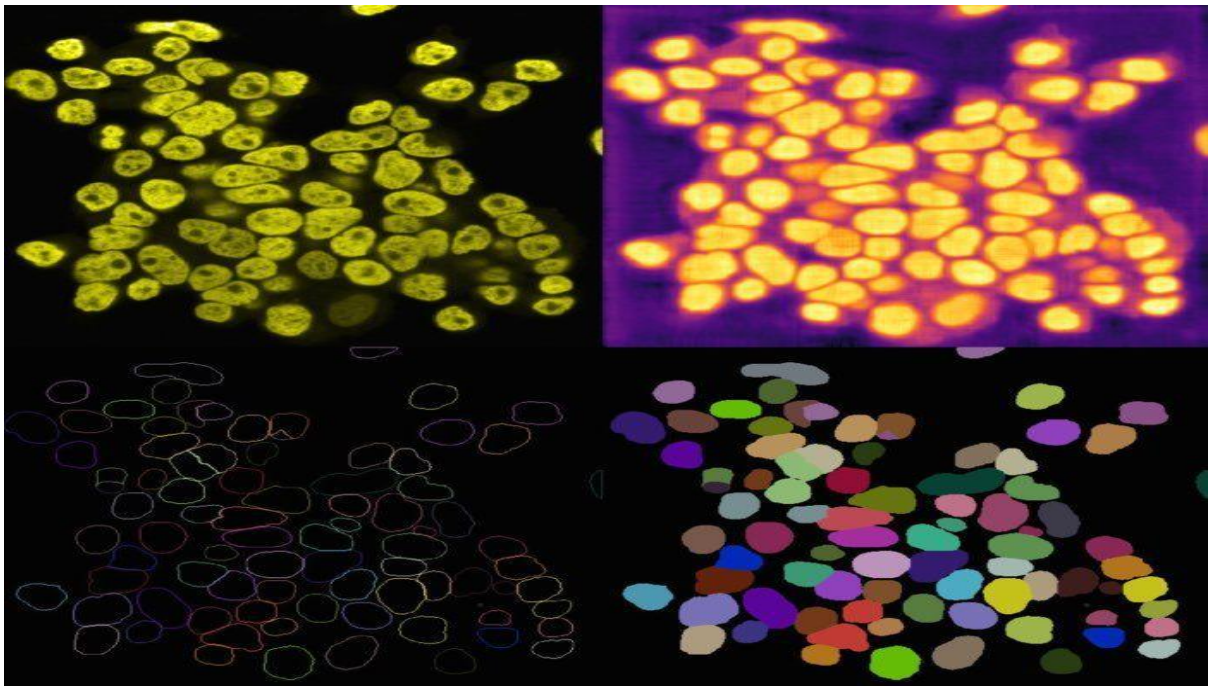
Decision: Marker-controlled watershed segmentation is useful when multiple cells touch and form a single connected component. Region growing is suitable when individual cells have relatively uniform intensity or colour.

Role: Region-based segmentation separates touching cells, reduces counting errors, and produces distinct cell regions for further analysis.

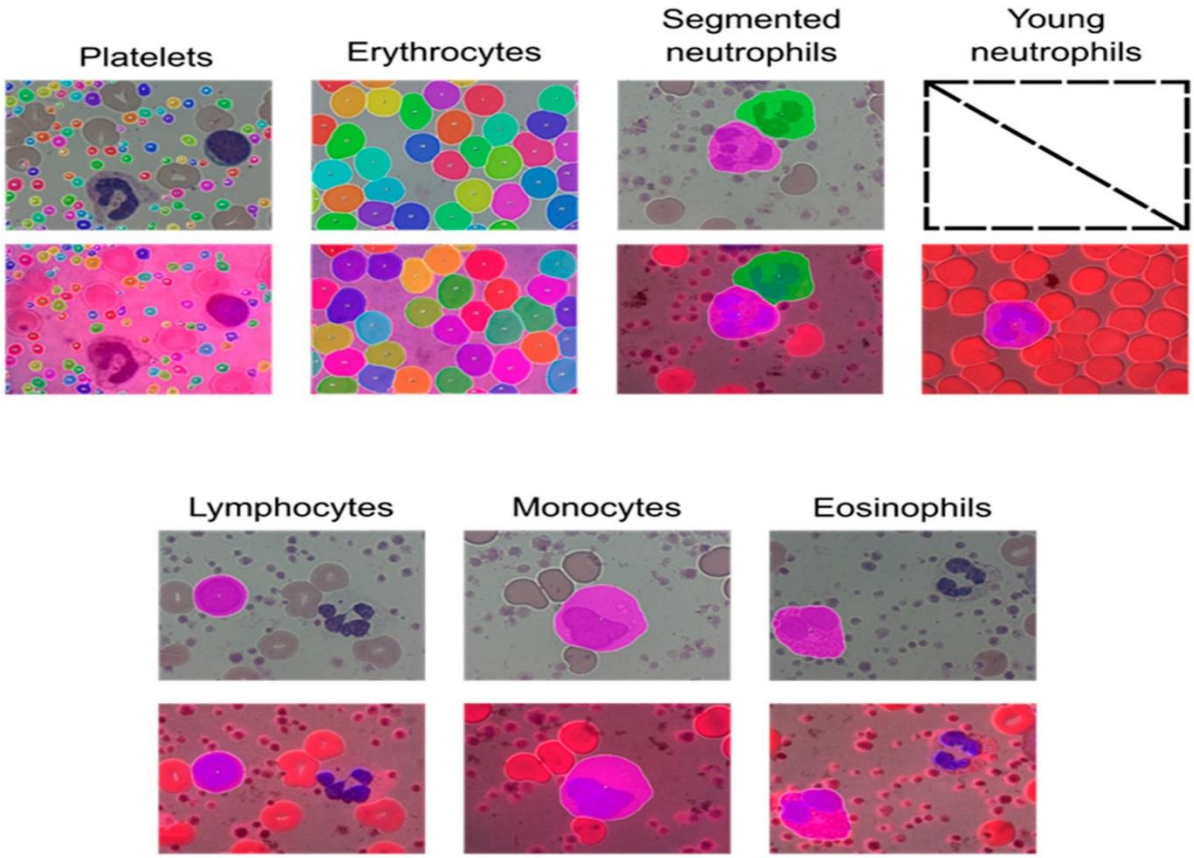
Diagram 4 – Region-Based Segmentation



Thresholded Image



Region Growing / Watershed



Separated Cells

8. Edge Detection and Boundary Detection

Edge detection identifies sharp changes in pixel intensity, while boundary detection extracts the outlines of individual blood cells. These techniques help determine cell shape, size, and position.

Techniques:

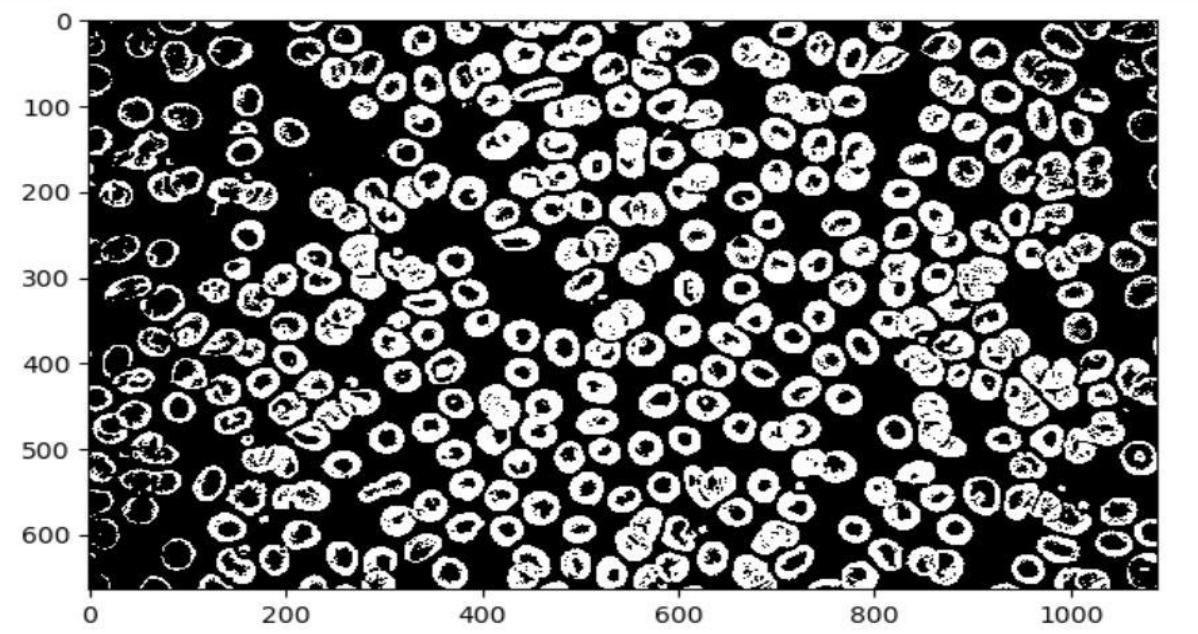
- **Sobel operator:** Calculates intensity gradients to identify possible cell edges**.**
- **Canny edge detector:** Detects strong edges while reducing false detections through smoothing and edge linking.
- **Contour extraction:** Traces the outlines of connected cell regions.
- **Morphological operations:** Opening removes small noise regions, while closing can fill small gaps in cell boundaries.

Procedure:

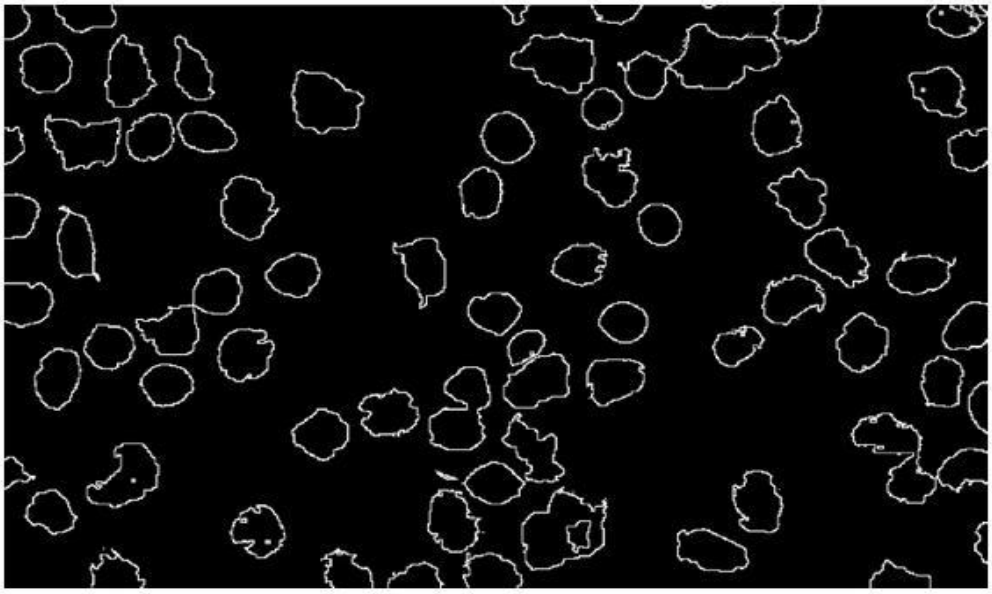
1. Apply an edge detector to the enhanced or segmented image.
2. Identify the edges corresponding to individual cells.
3. Extract closed contours around the segmented regions.
4. Overlay the detected contours on the original microscopic image.
5. Calculate the area, perimeter, shape, and position of each cell.

Role: Edge and boundary detection produces clear cell outlines and supports accurate cell counting and morphological analysis.

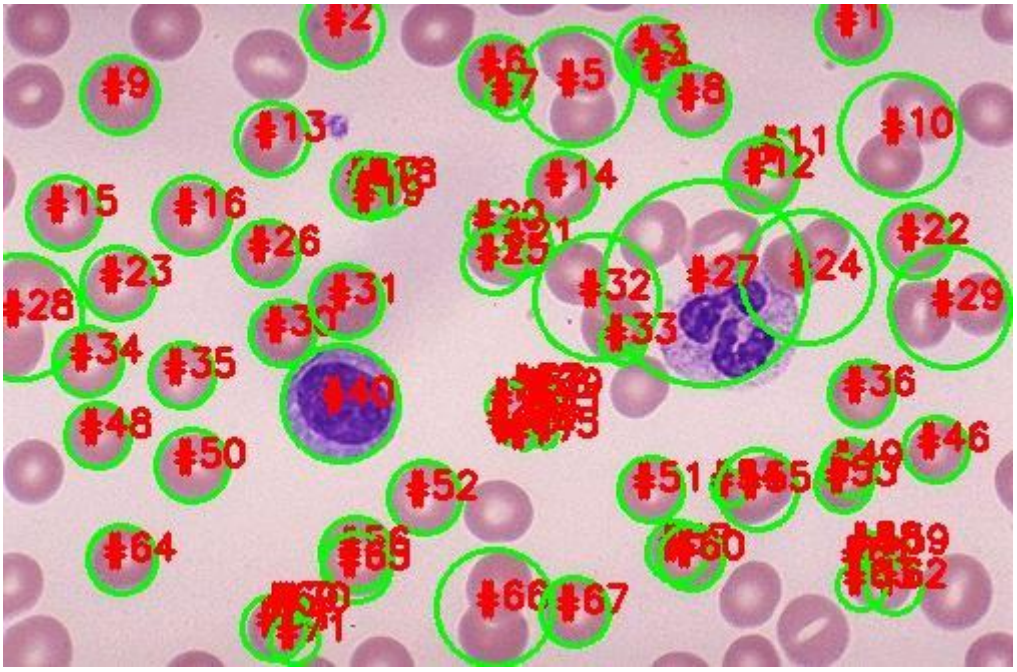
Diagram 5 – Edge and Boundary Detection



Segmented Image



Canny Edge Detection



Detected Boundaries

9. Final Segmentation Result

The final segmented image contains individual cell regions with visible boundaries. The results can be used to count red and white blood cells, estimate their sizes, and examine their shapes.

Processing stage	Expected result
Contrast enhancement	Improves cell visibility
Spatial filtering	Reduces unwanted noise
Thresholding	Separates cells from the background
Region-based segmentation	Separates touching cells
Edge detection	Identifies cell edges
Contour extraction	Outlines individual cells

Expected outcome: A clearer microscopic image containing candidate cell regions with identifiable boundaries, suitable for automated counting and further analysis.

10. Conclusion

Blood cell segmentation combines contrast enhancement, histogram processing, spatial filtering, thresholding, region-based segmentation, and edge detection to improve microscopic image analysis.

Histogram equalization and CLAHE improve the contrast between cells and the background, while median filtering removes unwanted noise without excessively blurring cell boundaries. Adaptive thresholding handles uneven staining, and watershed segmentation separates touching cells into individual regions. Finally, edge detection and contour extraction outline the cells for measurement and counting.

The resulting segmented image supports computer-assisted pathology by improving cell identification, counting, and shape analysis. However, the techniques must be validated on representative blood smear images to minimise false detections and incorrect separation.

5)Text Extraction from Degraded Historical Documents

1. Introduction

Historical document digitisation is an important application of computer vision used by archives, libraries, museums, and research institutions to preserve old handwritten manuscripts. Scanned documents are processed to improve readability and prepare the text for handwriting recognition.

However, historical manuscripts often contain yellowed paper, faded ink, stains, ink bleed-through from the reverse side, and uneven illumination. These problems reduce the contrast between handwritten strokes and the paper background, making accurate text extraction difficult.

Objective: To improve the contrast of faded handwriting, reduce stains and noise, strengthen pen strokes, separate text from the background, and detect text boundaries for reliable handwriting recognition.

2. Identification of Key Issues

- Yellowed paper: Age-related discoloration creates a non-uniform background.
- Faded ink: Weak pen strokes become difficult to distinguish from the paper.
- Ink bleed-through: Writing on the reverse side appears through the paper.
- Stains and noise: Dust, spots, and ink marks interfere with text extraction.
- Uneven illumination: Some parts of the scanned page appear brighter or darker.
- Broken strokes: Faint handwriting may contain gaps that affect recognition.
- Boundary errors: Background stains can be incorrectly identified as text.

Therefore, the system requires grey-level transformations, histogram processing, smoothing, sharpening, thresholding, and boundary detection.

3. Proposed Processing Pipeline

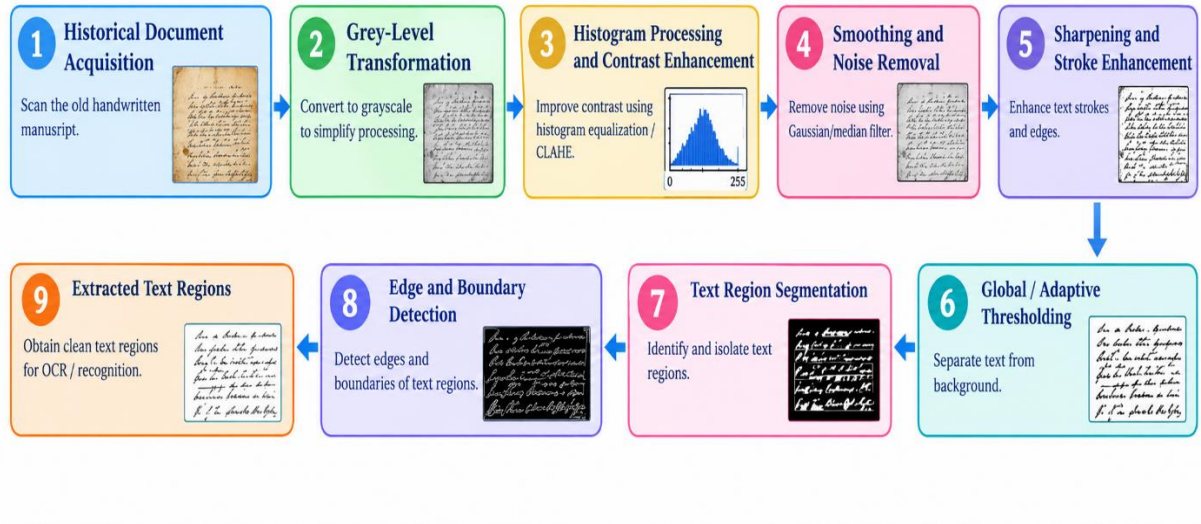


Figure 1: Historical Document Image Processing Flowchart.

Role: This pipeline enhances the scanned manuscript, separates handwritten text from the paper background, and produces clean text regions for handwriting recognition.

4. Grey-Level Transformations and Histogram Processing

Grey-level transformations modify pixel intensities to improve the visibility of faded handwritten strokes.

Techniques:

- Contrast stretching: Expands the intensity range to make faint writing more visible.
- Log transformation: Enhances details in darker intensity ranges.
- Gamma correction: Adjusts image brightness to reveal weak strokes.
- Histogram equalisation: Redistributes intensity values to improve overall contrast.
- CLAHE: Enhances local contrast while limiting excessive amplification of stains and noise.

A simple contrast-stretching transformation is:

$$g(x,y) = \frac{f(x,y) - f_{\min}}{f_{\max} - f_{\min}} \times 255$$

Where:

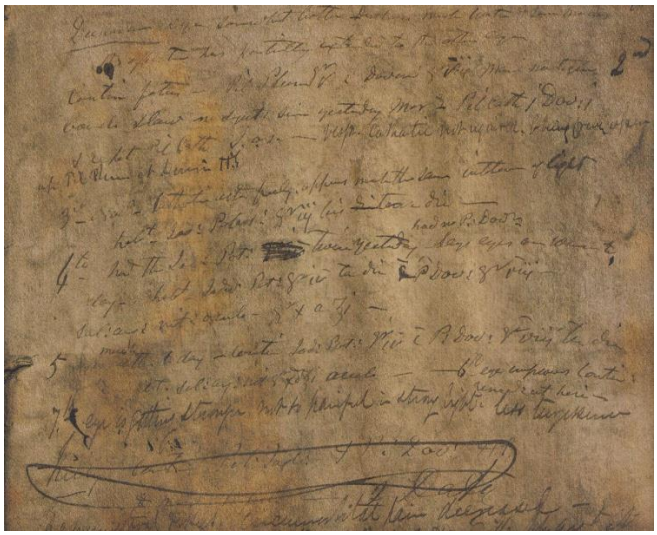
- f(x,y) is the original pixel intensity.
- $f_{\min}$ *is the minimum intensity.*

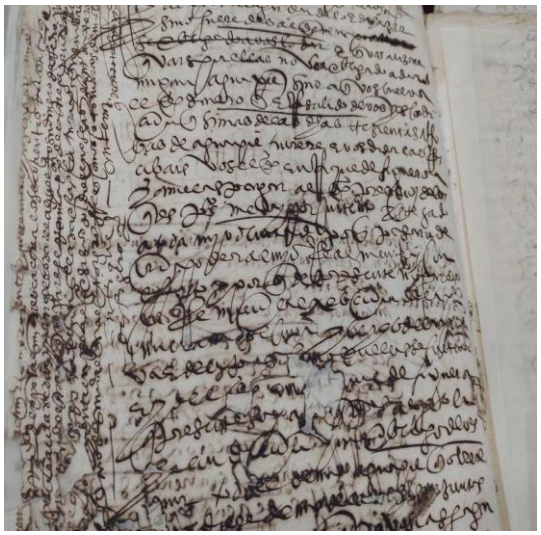
  $f_{\max}$ is the maximum intensity.

  g(x,y) is the enhanced pixel intensity.

Role: Grey-level transformations and histogram processing increase the difference between faded ink and the paper background, improving the visibility of handwritten characters.

Diagram 1 – Contrast Enhancement





Original Manuscript

Enhanced Manuscript

5. Smoothing and Sharpening Filters

Smoothing and sharpening are used together to reduce unwanted image variations while preserving the shape of handwritten strokes.

A. Smoothing filters

- **Median filter:** Removes isolated dark or bright spots while preserving many stroke edges.
- **Gaussian filter:** Reduces fine noise through weighted averaging, but excessive smoothing may erase faint handwriting.

- **Bilateral filter:** Reduces noise while preserving strong edges between strokes and the background.

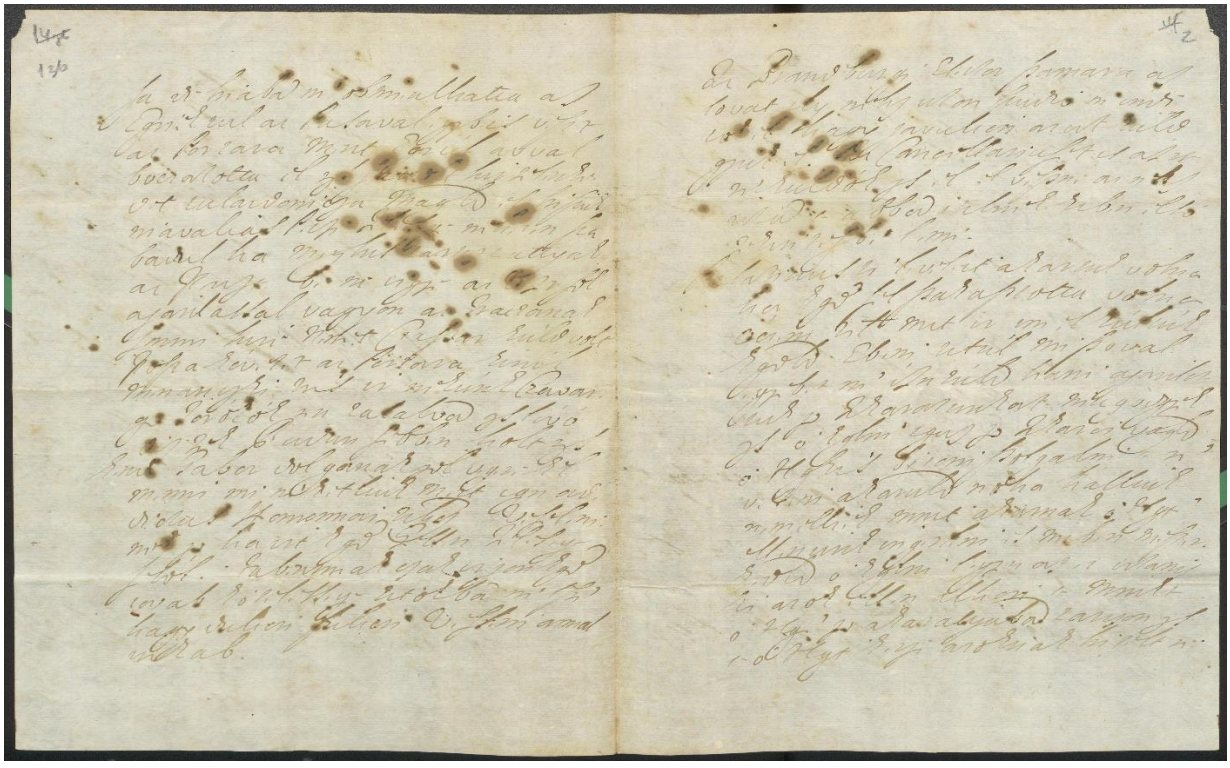
B. Sharpening filters

- **Laplacian filter:** Highlights rapid intensity changes and helps emphasise stroke boundaries.
- **Unsharp masking:** Subtracts a blurred version of the image from the original to enhance fine details.
- **High-pass filtering**: Emphasises rapid changes in intensity, making some pen strokes more prominent.

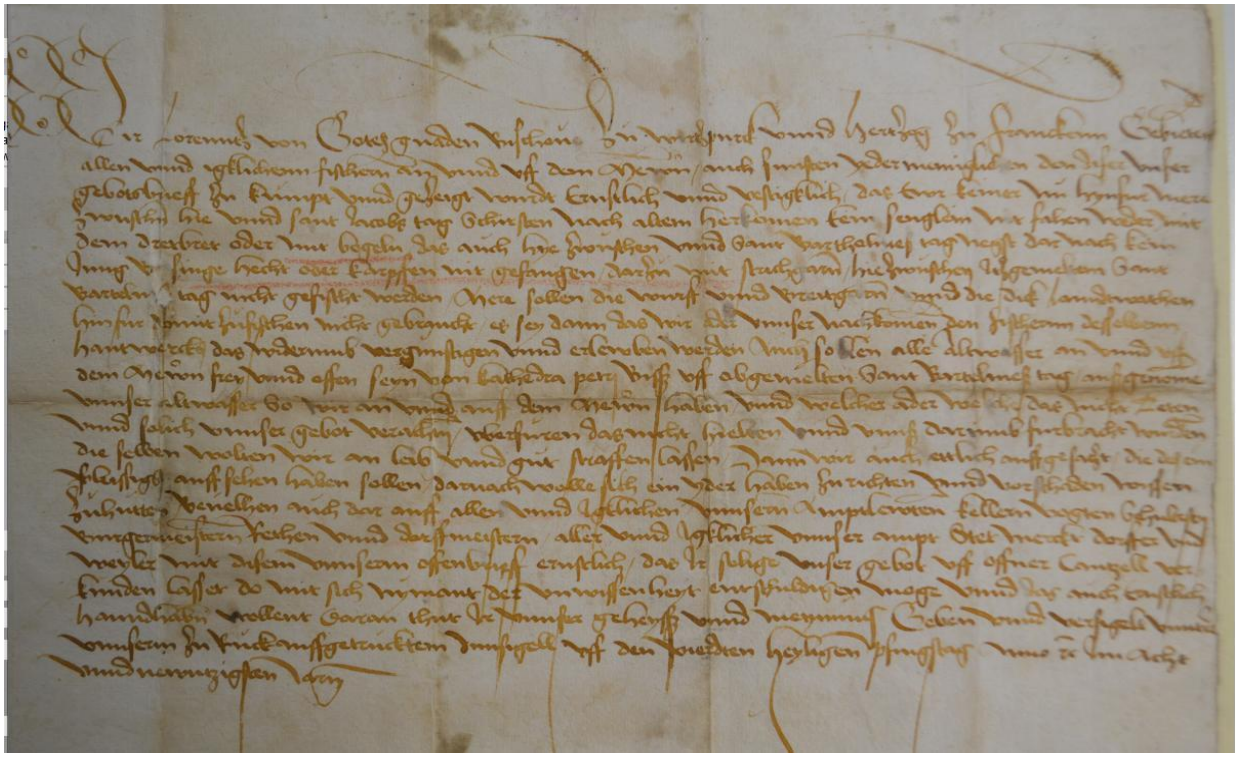
Decision: Apply mild median or bilateral filtering first to reduce stains and small noise. Then use controlled sharpening to strengthen faded strokes. Excessive smoothing or sharpening should be avoided because it can destroy handwriting details or amplify ink bleed-through.

Role: Smoothing reduces unwanted noise, while sharpening improves the visibility of character strokes and prepares the image for thresholding.

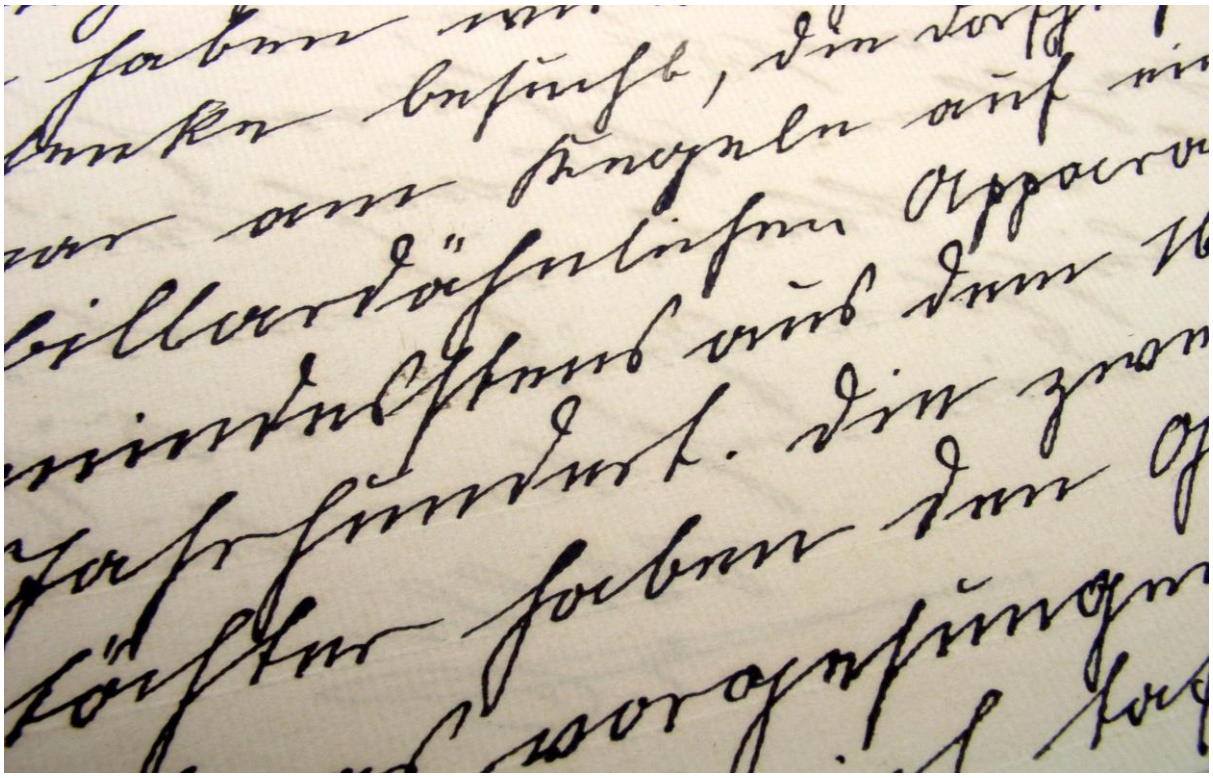
Diagram 2 – Smoothing and Sharpening



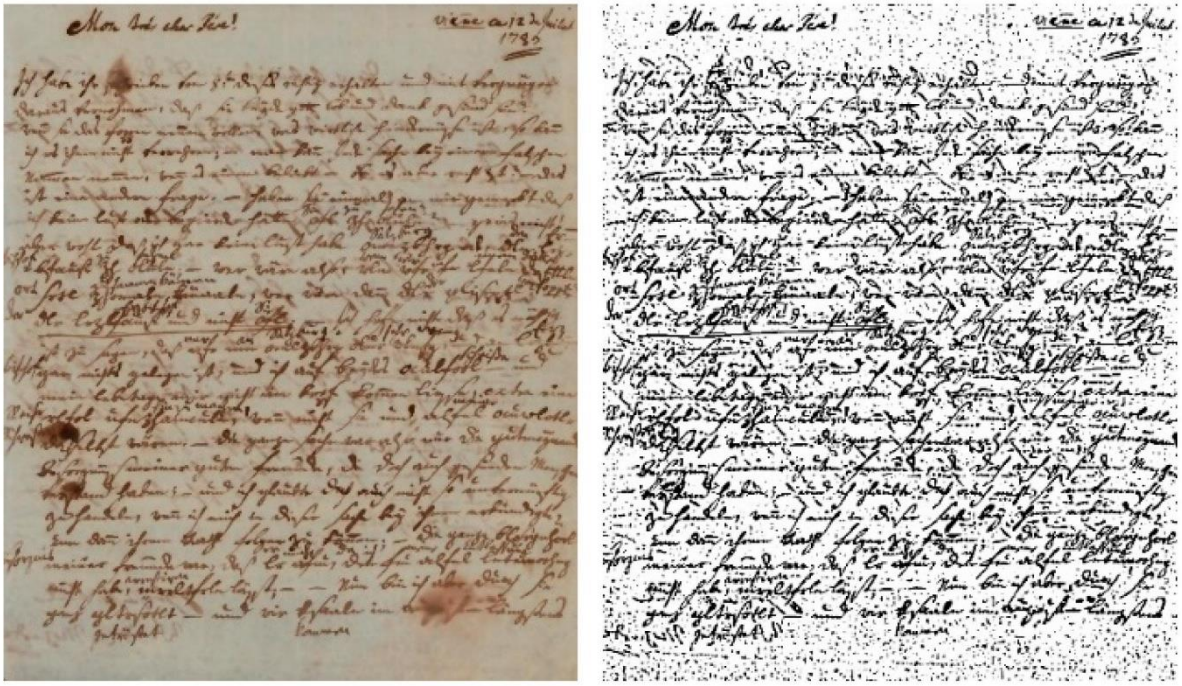
Noisy Manuscript



After Smoothing



Sharpened Strokes



(d)

Ready for Thresholding

6**. Global and Adaptive Thresholding**

Thresholding converts a grayscale manuscript into a binary image containing text and background pixels.

$$g(x,y) = \begin{cases} 0, & f(x,y) < T \\ 255, & \text{otherwise} \end{cases}$$

Here, f(x,y) s the input intensity, T is the threshold value, and g(x,y) is the binary output. This convention assumes dark ink on a light background.

Global thresholding – Otsu's method

- Selects a single threshold for the entire document.
- Works well when the paper background is relatively uniform.
- May fail when the page contains shadows, stains, or varying brightness.

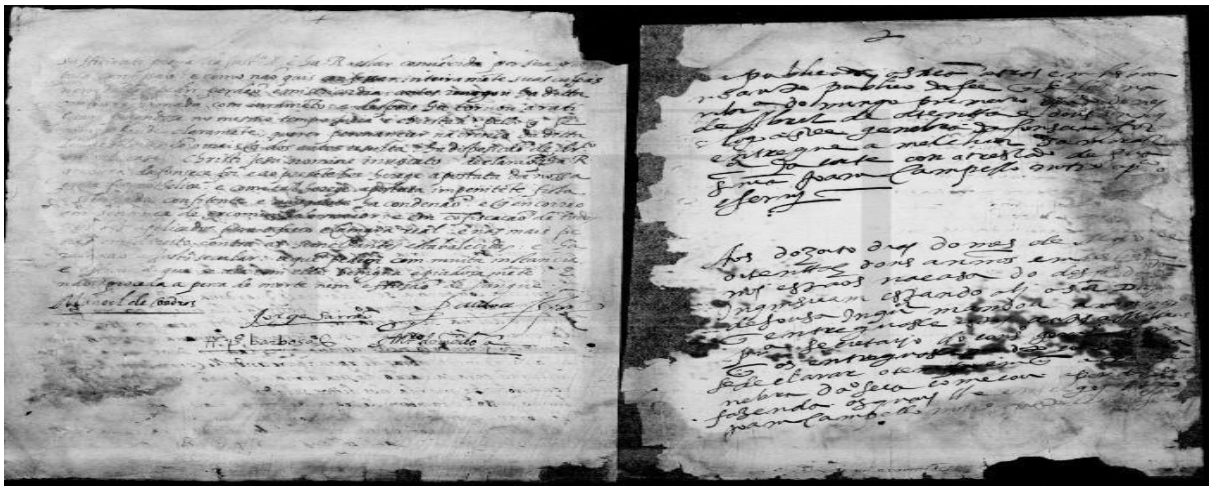
Adaptive thresholding

- Calculates a local threshold for each neighbourhood.
- Handles uneven illumination and local paper discoloration.
- Usually works better when the manuscript contains faded strokes and background variations.

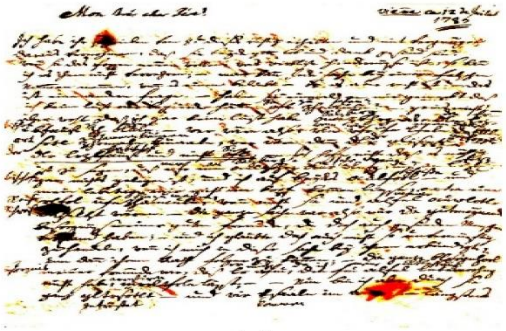
Decision: Adaptive thresholding is the preferred approach for heavily degraded manuscripts because it handles local changes in illumination and background intensity. Otsu's method is suitable when the document has relatively uniform illumination and good contrast.

Role: Thresholding separates handwritten text from the background and produces a binary image for further processing.

Diagram 3 – Thresholding



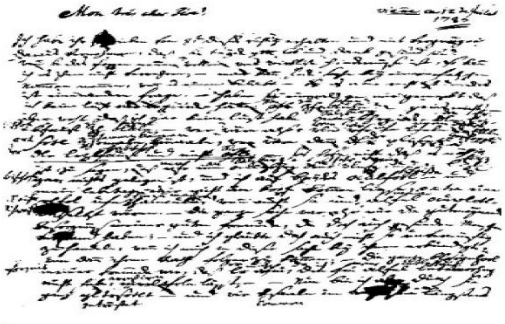
Grayscale Manuscript



(a)



(b)



(c)

Binary Text Image

7. Text Region Segmentation

After thresholding, text region segmentation identifies the locations of individual words, lines, and paragraphs.

Techniques:

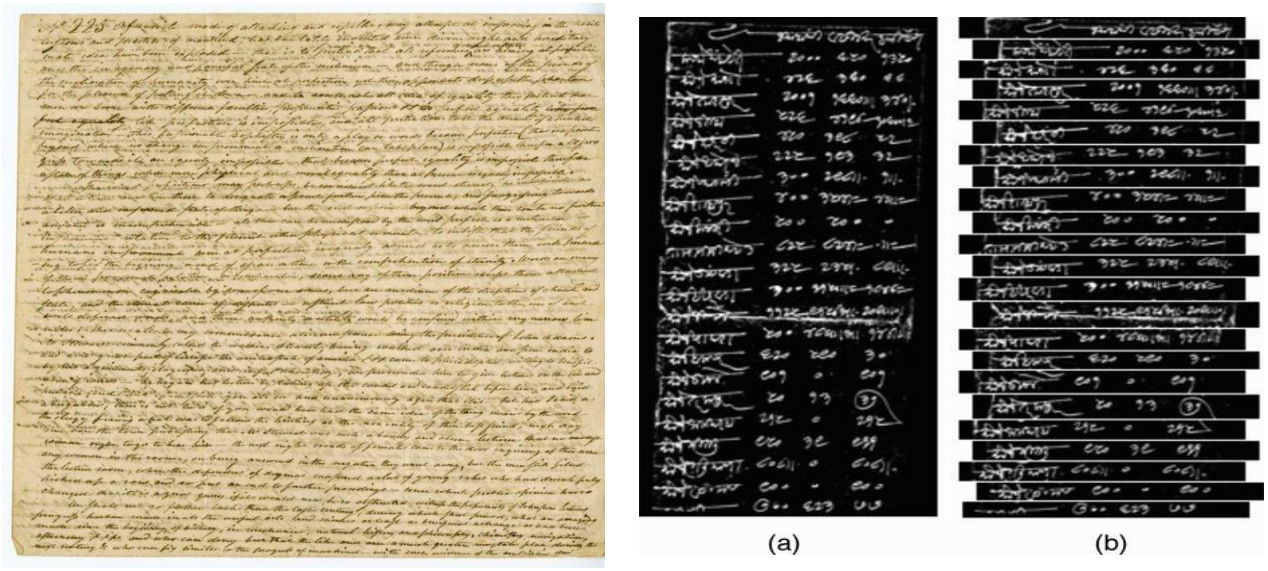
- Connected-component analysis: Groups connected foreground pixels into separate text components.
- Morphological opening: Removes small isolated noise components.
- Morphological closing: Connects small gaps in strokes when appropriate.
- Line segmentation: Identifies horizontal text lines using projections or connected components.
- Region grouping: Groups characters and words according to their spatial relationships.

Procedure:

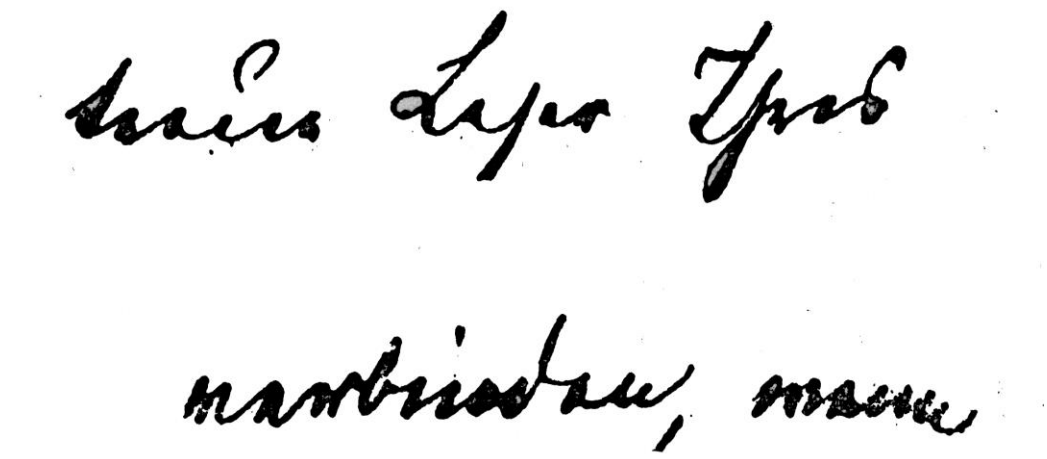
1. Analyse the binary image to identify foreground pixels.
2. Remove small isolated components caused by stains and dust.
3. Group connected components into characters and words.
4. Identify text lines using their vertical positions.
5. Extract the required text regions for handwriting recognition.

Role: Text region segmentation separates meaningful handwriting from the surrounding paper and prepares the manuscript for recognition systems.

Diagram 4 – Text Region Segmentation



Scanned Page **Text Lines**



Extracted Region

8. Edge Detection and Boundary Detection

Edge detection identifies rapid changes in intensity, while boundary detection outlines the regions containing handwritten text.

Techniques:

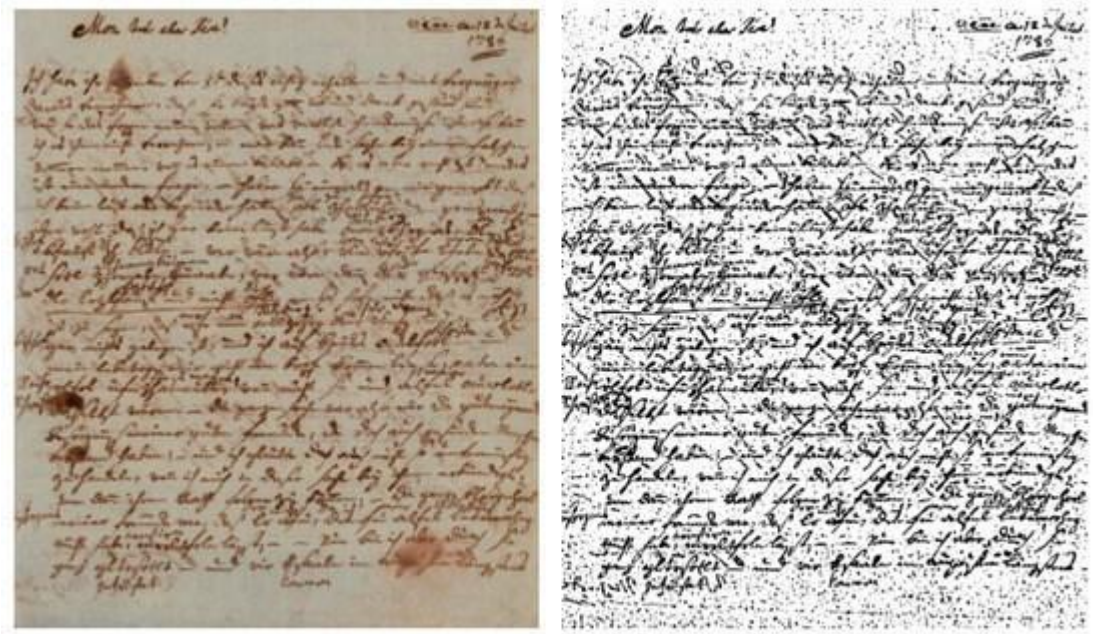
- **Sobel operator**: Detects intensity gradients around handwriting strokes.
- **Canny edge detector:** Identifies prominent edges while reducing false detections caused by noise.
- **Contour extraction:** Traces connected text components and their boundaries.
- **Morphological processing**: Helps repair small gaps and remove unwanted components.

Procedure:

1. Apply edge detection to the enhanced or binary document.
2. Identify the boundaries of handwritten strokes and text components.
3. Extract contours around individual text regions.
4. Reject obvious noise components based on size and shape.
5. Overlay the detected boundaries on the scanned page.

Role: Boundary detection identifies the location and shape of text regions, making it easier to isolate handwriting before recognition.

Diagram 5 – Edge and Boundary Detection

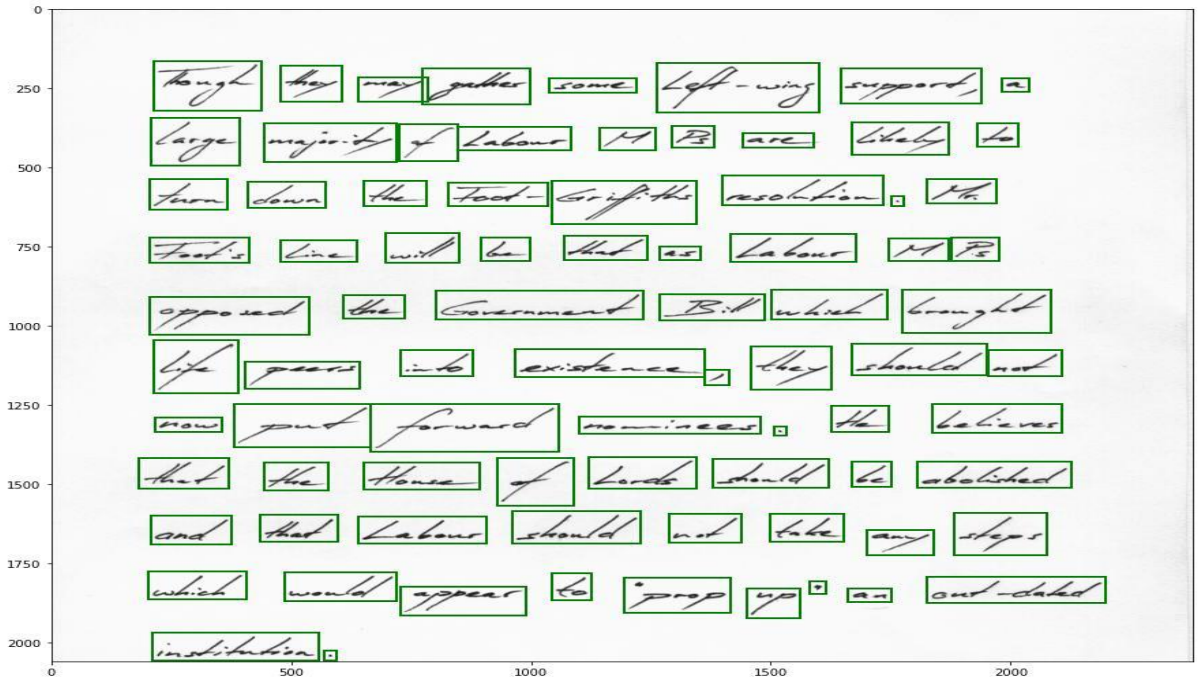


(d)

Binary Text



Detected Edges



Text Boundaries

9. Expected Results

Processing stage	Expected result
Grey-level transformations	Improves visibility of faded ink
Histogram processing	Enhances text-background contrast
Smoothing filters	Reduces stains and isolated noise
Sharpening filters	Strengthens visible pen strokes
Adaptive thresholding	Separates text under uneven illumination
Text segmentation	Extracts lines, words, and text regions
Boundary detection	Identifies the outlines of text regions

Expected outcome: A cleaner binary document with more visible handwriting, fewer unwanted background components, and clearly identified text regions suitable for handwriting recognition.

10. Conclusion

Text extraction from degraded historical documents combines grey-level transformations, histogram processing, smoothing, sharpening, thresholding, text segmentation, and boundary detection. Contrast stretching and CLAHE improve faded handwriting, while median or bilateral filtering reduces noise and preserves important stroke details. Controlled sharpening strengthens visible pen strokes.

Adaptive thresholding is particularly useful for historical pages with yellowed paper, stains, and uneven illumination. Connected-component analysis and morphological operations help separate text from the background, while edge detection and contour extraction identify text boundaries.

The final output is a cleaned and segmented document that can be passed to a handwriting recognition system. The processing parameters must be carefully selected to preserve genuine strokes and avoid treating bleed-through or stains as actual text.